Chapter 31: Information Retrieval

Database System Concepts, 7th Ed.

Introduction to Information Retrieval

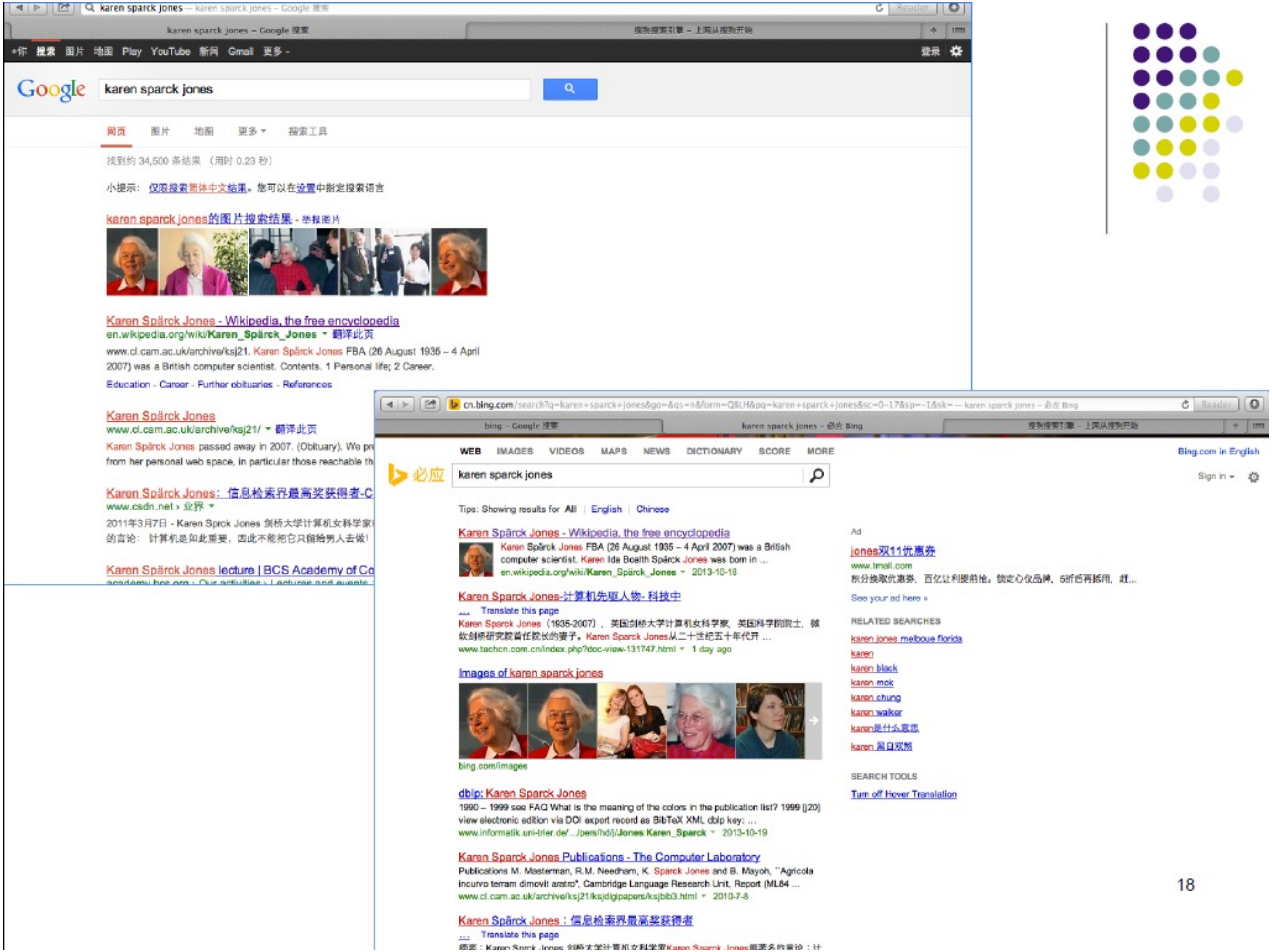
Outline

- What is Information Retrieval ?
- Basic IR concepts and models
  - Boolean model
  - Vector space model
  - Probabilistic model
  - Language model

Some of the slides are adapted from the publically available ppt for the textbook "Introduction to Information Retrieval"

Outline

- **What is Information Retrieval ?**
- Basic IR concepts and models



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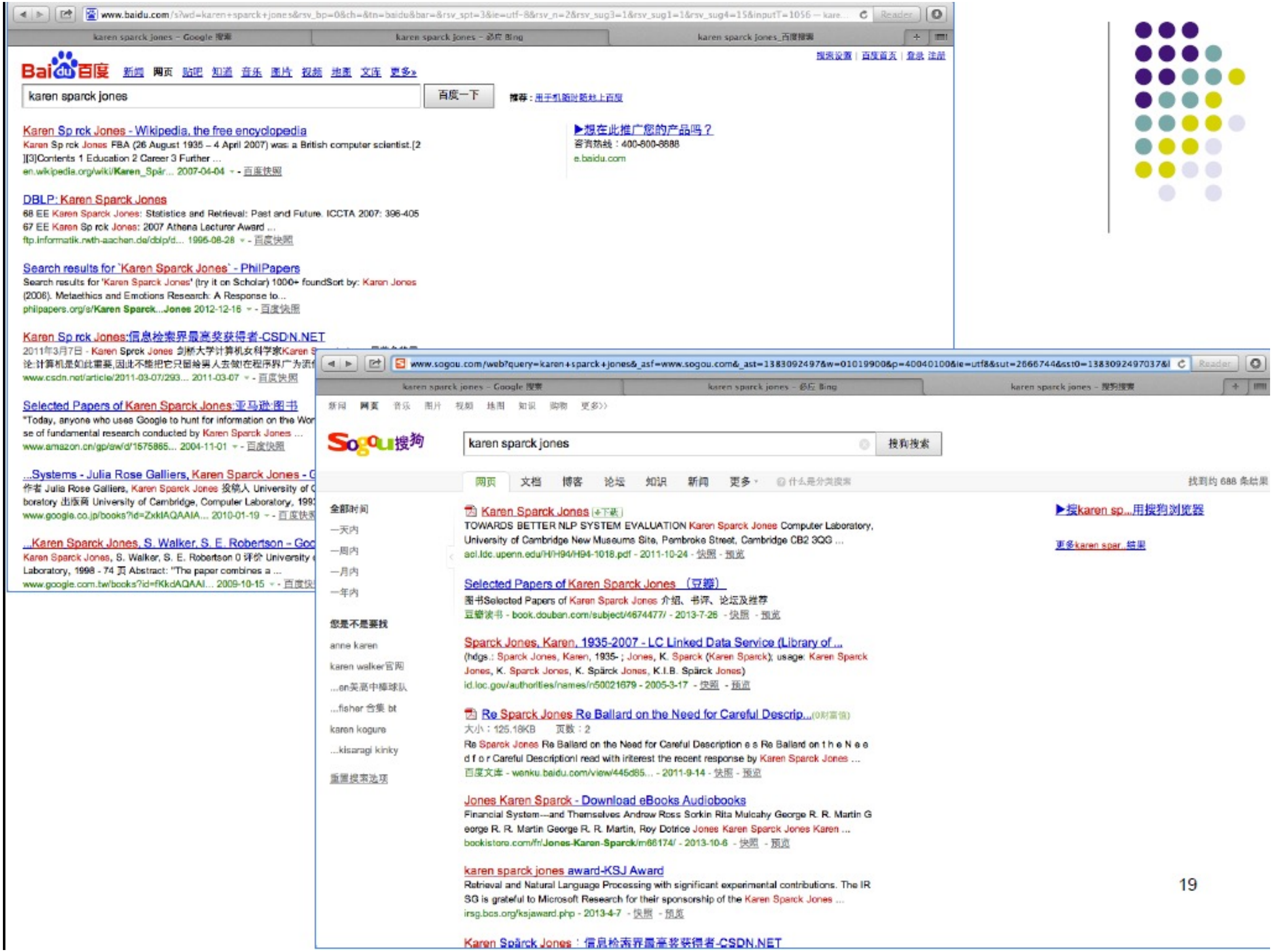
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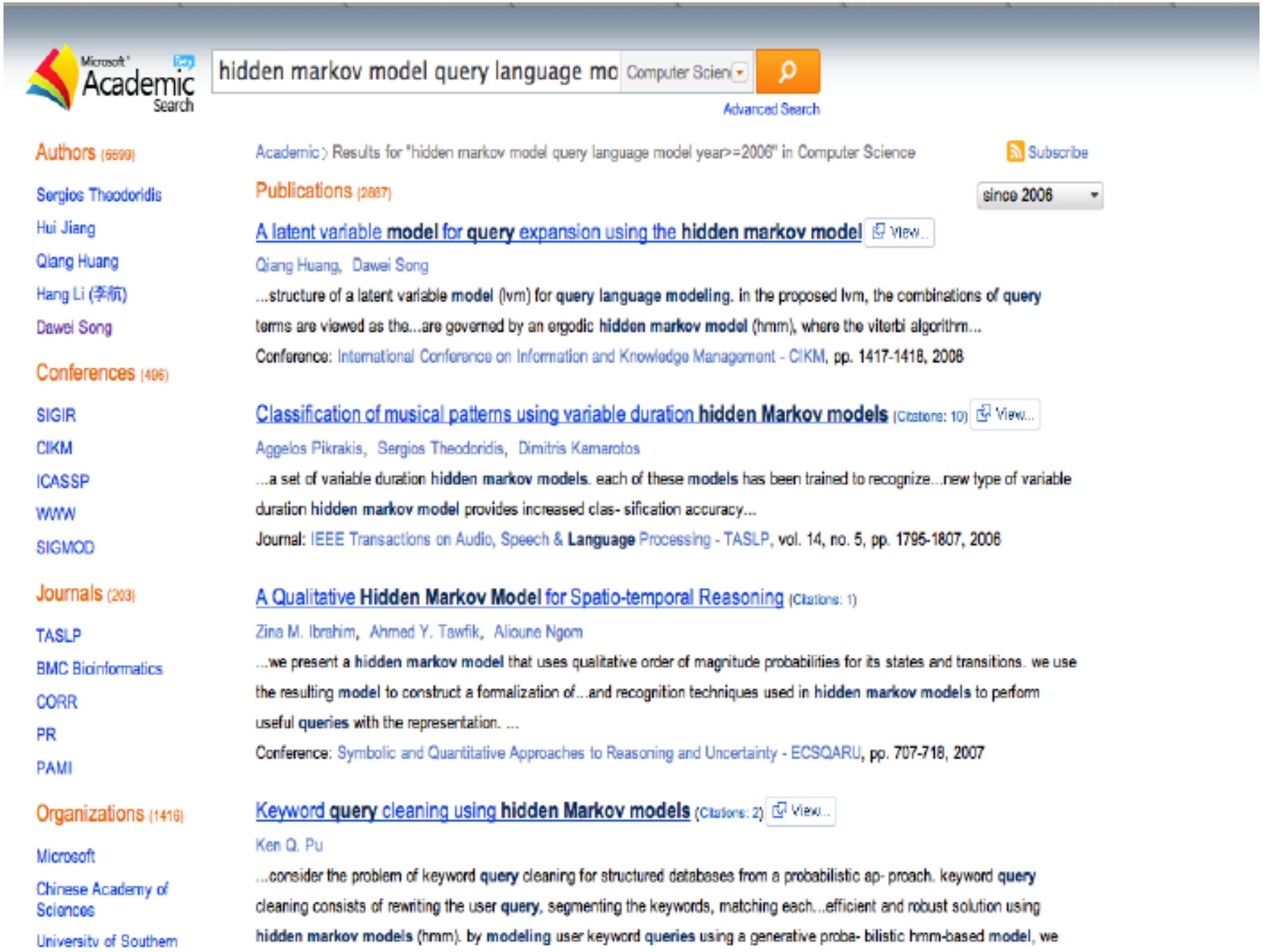

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A Qualitative Hidden Markov Model for Spatio-temporal Reasoning (Citations: 1)
Zina M. Ibrahim, Ahmed Y. Tawfik, Alioune Ngom
...we present a hidden markov model that uses qualitative order of magnitude probabilities for its states and transitions. we use the resulting model to construct a formalization of...and recognition techniques used in hidden markov models to perform useful queries with the representation. ...
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Keyword query cleaning using hidden Markov models (Citations: 2)
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...consider the problem of keyword query cleaning for structured databases from a probabilistic ap- proach. keyword query cleaning consists of rewriting the user query, segmenting the keywords, matching each...efficient and robust solution using hidden markov models (hmm). by modeling user keyword queries using a generative proba- bilistic hmm-based model, we


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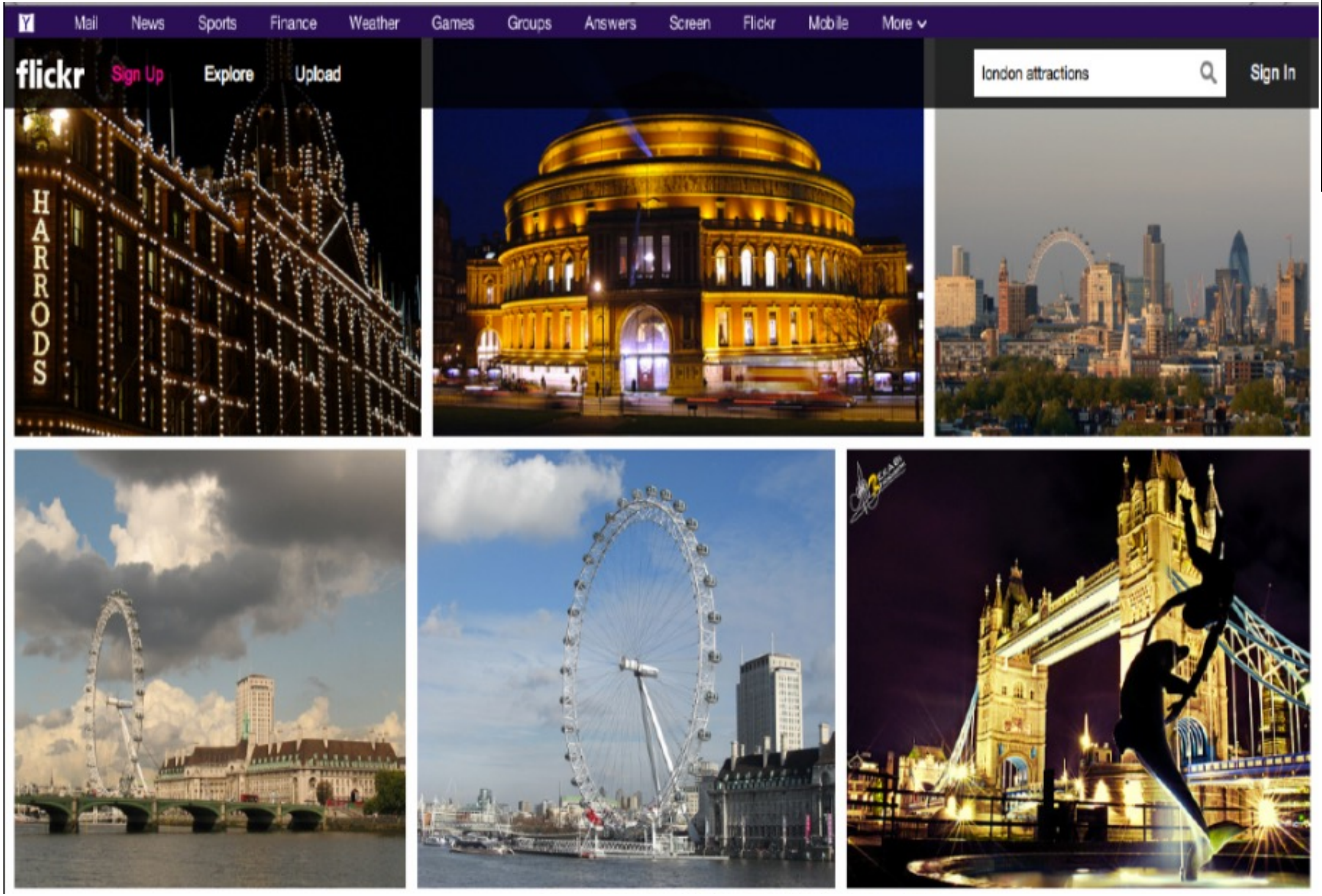

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Common features of information retrieval applications

- **Goal:** To retrieve information that is **relevant** to the user's **information need** and help the user complete an information seeking **task**

  Information need:
  books, personal info, products, news

Information Retrieval

- The subject that aims at seeking desired information given a user's information need. It involves information acquisition, storage, organization and access.

- The process of finding material of an unstructured nature (usually text or multimedia) that satisfies an information need from within large collections (usually stored on computers).

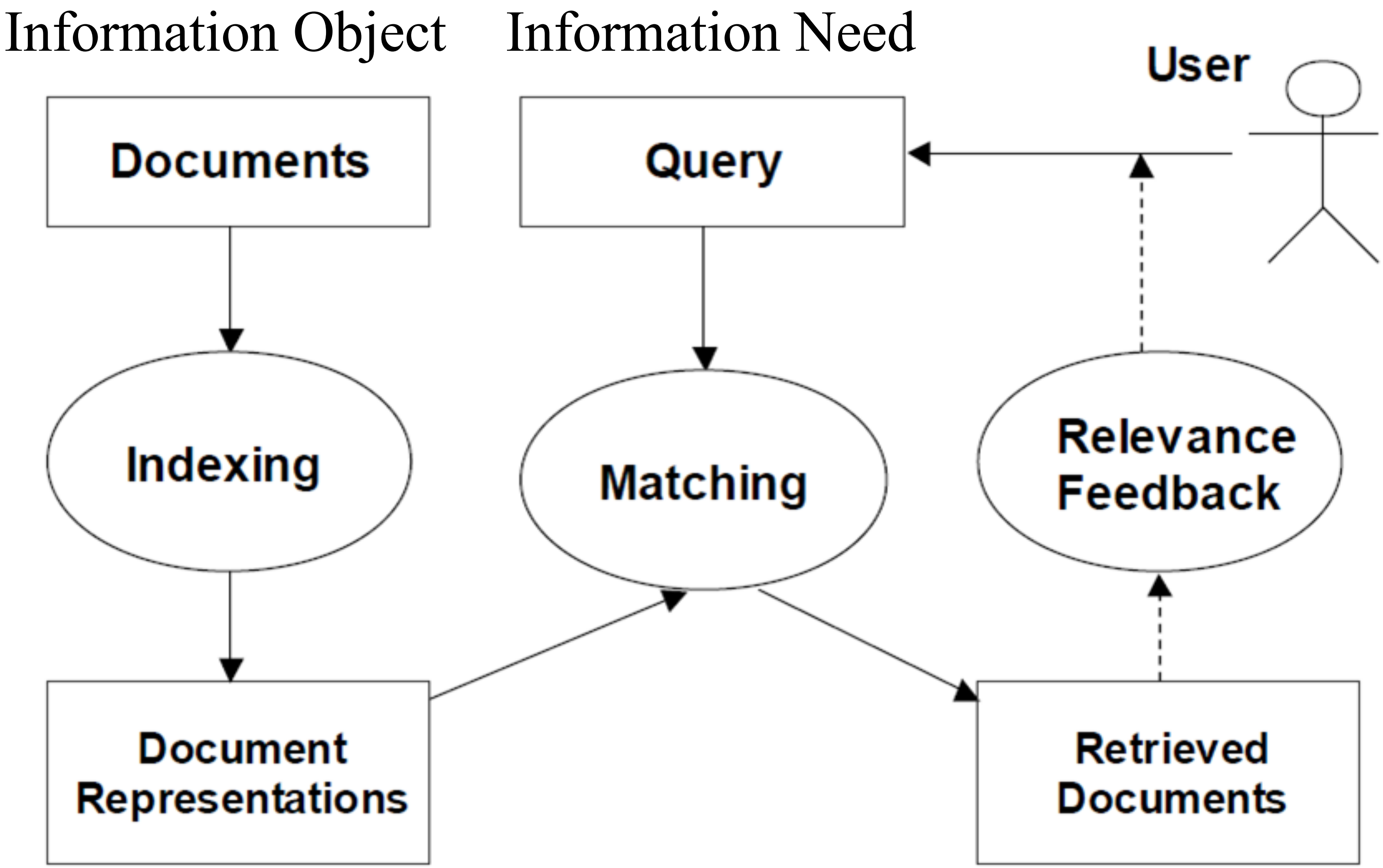
Information Retrieval

- How to find relevant objects?

Define and calculate the "similarity" function:

- Give the information need: **query**
- Objects to be retrieved: **text or multimedia**
- Match between query and object: **relevance score**

The classical search model



IR is also science

- IR is not merely an engineering problem, but also a scientific subject
  - NLP
  - Pattern recognition,
  - Artificial intelligence
  - Machine learning
  - Human-machine interaction
  - User behavior and cognition
  - Modeling complex information objects
  - Social networks
  - Big data
  - ……

The complex interactions among above topics

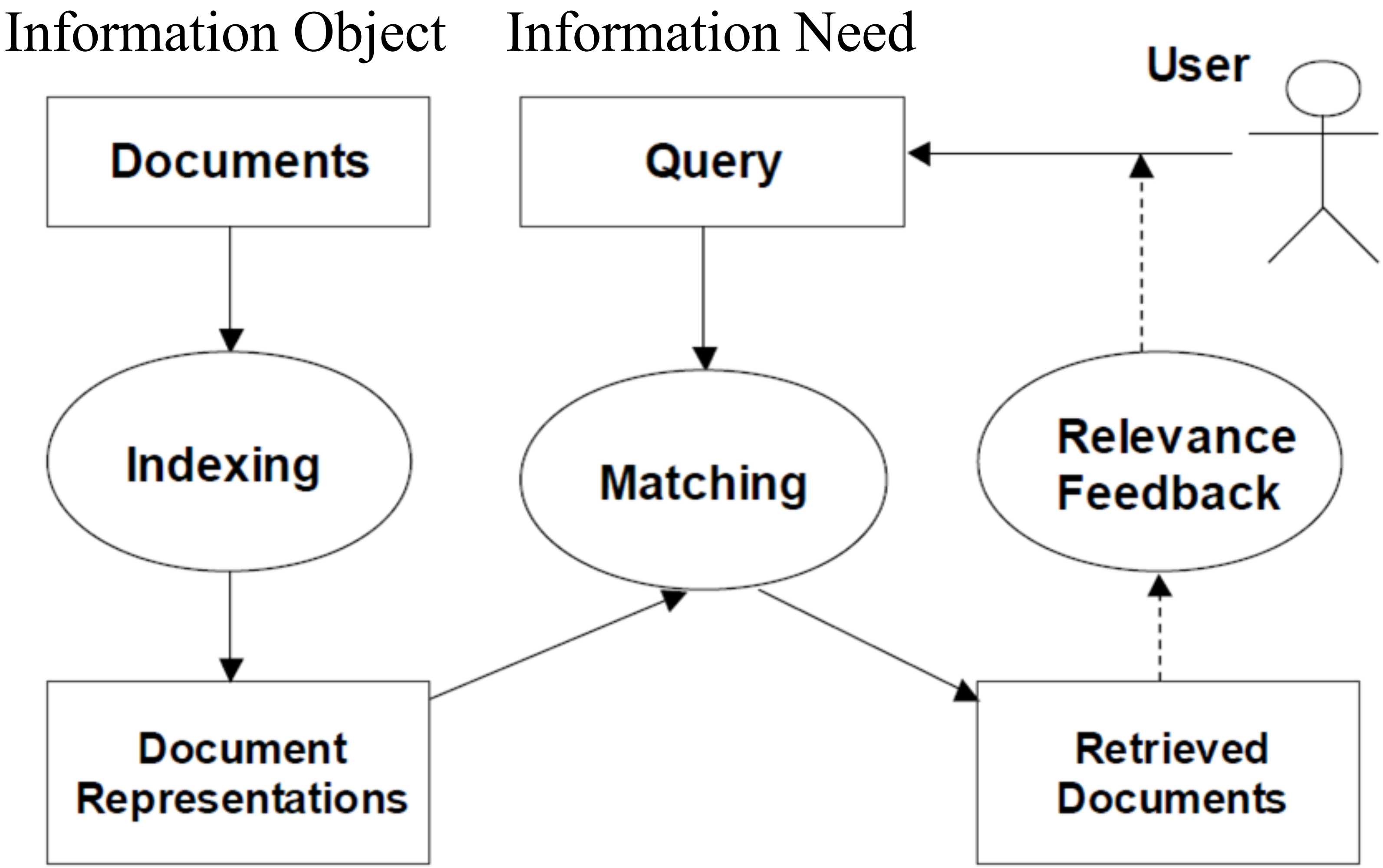
IR Applications

- Web search engines (e.g., Google, Baidu, Bing, Sougou)
- Intranet and domain-specific search systems
- E-commerce (e.g., Alibaba, Amazon)
- Social networking site (e.g., Weibo, Facebook, Twitter)
- Intelligence and security
- Mobile apps
- Digital libraries
- Consumer behavior analysis
- News/Ads push
- Question-Answering system
- ……

Outline

- What is Information Retrieval ?
- **Basic IR concepts and models**

The classic search model



Basic IR concepts

- The process of finding material of an unstructured nature (usually text or multimedia) that satisfies an information need within large collections (usually stored on computers).

- Unstructured
- Documents
- Information Need
- Collection

IR vs. databases: Structured vs unstructured data

- Structured data tends to refer to information in "tables"

Employee	Manager	Salary
Smith	Jones	50000
Chang	Smith	60000
Ivy	Smith	50000

Typically allows numerical range and exact match (for text) queries, e.g.,

Salary < 60000 AND Manager = Smith.

IR vs. databases

- IR systems don’t deal with transactional updates (including concurrency control and recovery)
- Database systems deal with structured data, with schemas that define the data organization
- IR systems deal with some querying issues not generally addressed by database systems
  - Approximate searching by keywords
  - Ranking of retrieved answers by estimated degree of relevance

Unstructured data

- Typically refers to free text
- Allows
  - Keyword queries including operators
  - More sophisticated queries e.g.,
    - find all web pages dealing with *drug abuse*
- Classic models for searching textual documents

Semi-structured data

- In fact almost no data is "unstructured"

  <title>homepage</title>

  <body>…</body>

- Facilitates "semi-structured" search such as
  - *Title* contains data AND *Bullets* contain search
    - other structures not mentioned here, such as linguistics and Hyperlinks

Assumptions of classical IR

- **Collection:** A set of documents
  - Assume it is a static collection for the moment
- **Goal:** Retrieve documents with information that is **relevant** to the user's **information need** and helps the user complete a **task**
- **Relevance:**
  - Objective
  - The degree of matching
  - Depend on specific applications

Term-document incidence matrices

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	1	1	0	0	0	1
Brutus	1	1	0	1	0	0
Caesar	1	1	0	1	1	1
Calpurnia	0	1	0	0	0	0
Cleopatra	1	0	0	0	0	0
mercy	1	0	1	1	1	1
worser	1	0	1	1	1	0

Brutus AND Caesar BUT NOT Calpurnia

1 if play contains word, 0 otherwise

Bigger collections

- Consider N = 1 million documents, each with about 1000 words.

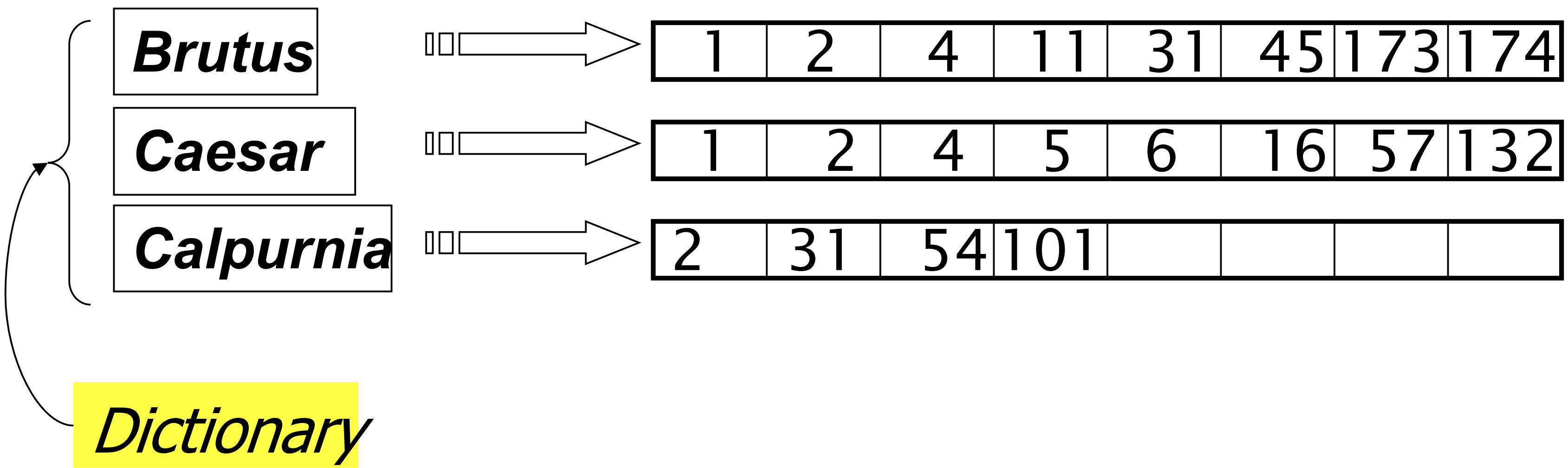
- Say there are M = 500K *distinct* terms among these.

Can't build the matrix

- 500K x 1M matrix has half-a-trillion 0's and 1's.
- But it has no more than one billion 1's.
  - matrix is extremely sparse.
- What's a better representation?
  - We only record the 1 positions.

Inverted index

- For each term *t*, we must store a list of all documents that contain *t*.
  - Identify each doc by a **docID**, a document serial number



Indexer steps: Token sequence

- **Sequence of (Modified token, Document ID) pairs.**

Doc 1

I did enact Julius Caesar I was killed i' the Capitol; Brutus killed me.

Doc 2

So let it be with Caesar. The noble Brutus hath told you Caesar was ambitious

Term	docID
I	1
did	1
enact	1
julius	1
caesar	1
I	1
was	1
killed	1
i'	1
the	1
capitol	1
brutus	1
killed	1
me	1
so	2
let	2
it	2
be	2
with	2
caesar	2
the	2
noble	2
brutus	2
hath	2
told	2
you	2
caesar	2
was	2
ambitious	2

Indexer steps: Dictionary & Postings

- Multiple term entries in a single document are merged.
- Split into Dictionary and Postings
- Doc. frequency information is added.

Core indexing step

Term	docID
ambitious	2
be	2
brutus	1
brutus	2
capitol	1
caesar	1
caesar	2
caesar	2
did	1
enact	1
hath	1
I	1
I	1
i'	1
it	2
julius	1
killed	1
killed	1
let	2
me	1
noble	2
so	2
the	1
the	2
told	2
you	2
was	1
was	2
with	2

term	doc. freq.	→	postings lists
ambitious	1	→	2
be	1	→	2
brutus	2	→	1 → 2
capitol	1	→	1
caesar	2	→	1 → 2
did	1	→	1
enact	1	→	1
hath	1	→	2
i	1	→	1
i'	1	→	1
it	1	→	2
julius	1	→	1
killed	1	→	1
let	1	→	2
me	1	→	1
noble	1	→	2
so	1	→	2
the	2	→	1 → 2
told	1	→	2
you	1	→	2
was	2	→	1 → 2
with	1	→	2

IR models

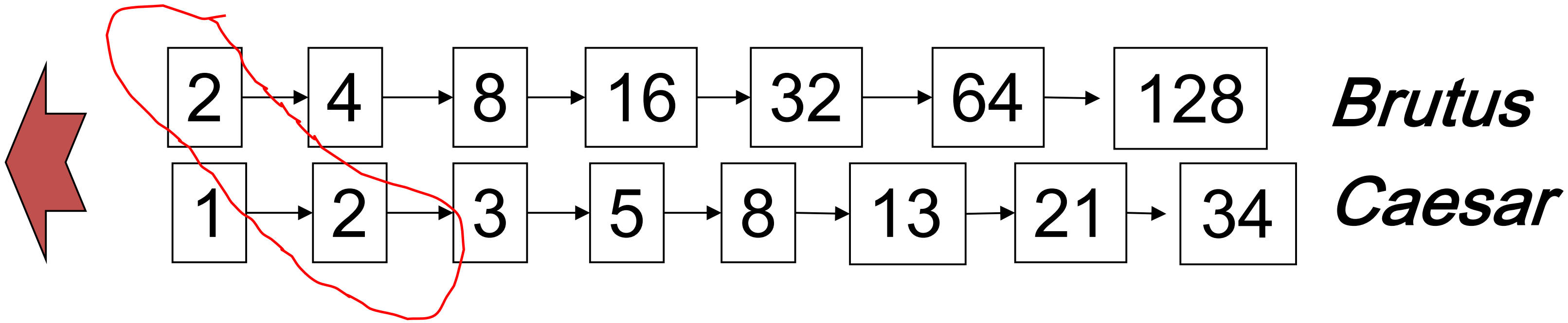
- Assume the inverted index has been build up.
- How do we process a query to match with documents to produce the relevance score?
  - Boolean model
  - Vector space model
  - Probabilistic model
  - Language model
  - ……

Boolean retrieval model

- Consider processing the query (using AND):

  Brutus *AND* ***Caesar***

  - Locate ***Brutus*** in the Dictionary;
    - Retrieve its postings.
  - Locate ***Caesar*** in the Dictionary;
    - Retrieve its postings.
  - "Merge" the two postings (intersect the document sets):



Other Boolean expressions

- **OR:** Brutus OR Caesar
  - The union of two inverted doc sets
- **NOT:** Brutus AND NOT Caesar
  - The difference of two inverted doc sets
- General Boolean expressions
  - (Brutus OR Caesar) AND NOT (Antony OR Cleopatra)

Advantages of Boolean retrieval model

- Easy to establish, perhaps the simplest model to build an IR system upon
  - Primary commercial retrieval tool for 3 decades
  - Many professional searchers still like Boolean search – e.g., medical search
    - You know exactly what you are getting
  - But that doesn't mean it actually works better

Problem with Boolean search

- Boolean expression can be very complex
  - It needs a lot of skills to come up with a query that produces a manageable number of hits.
  - AND gives too few; OR gives too many
  - Only good for expert users with precise understanding of their needs and the collection.
- It does not use the frequency of terms
- Documents either match or don't. It can not order over the documents in the collection for a query

Ranked retrieval models

- In **ranked retrieval**, the system returns an ordering over the (top) documents in the collection for a query
- Produces a manageable number of hits
- **Free text queries:** Rather than a query language of operators and expressions, the user's query is just one or more words in a human language
- **Precondition:** The ranking algorithm is valid, i.e., more relevant documents are ranked before less relevant ones.

Scoring as the basis of ranked retrieval

- We wish to return in order the documents most likely to be useful to the searcher
  - How can we rank-order the documents in the collection with respect to a query?
- Assign a score: say in [0, 1], to each document
  - This score measures how well document and query "match".

Query-document matching scores

- We need a way of assigning a score to a query-document pair
  - Let's start with a one-term query
  - If the query term does not occur in the document: score should be 0
  - The more frequent the query term in the document, the higher the score (should be)
- We will look at a number of alternatives for this.

Jaccard coefficient

- A commonly used measure of overlap of two sets *A* and *B*
  - jaccard$(A,B) = |A \cap B| / |A \cup B|$
  - jaccard$(A,A) = 1$
  - jaccard$(A,B) = 0$ if $A \cap B = 0$
- *A* and *B* don't have to be the same size.
- Always assigns a number between 0 and 1.

Jaccard coefficient: Scoring example

- Query: *ides of march*
- Document 1: *caesar died in march*
- Document 2: *the long march*

- *Jaccard(q,d1) = 1/6*

Issues with Jaccard for scoring

- It doesn't consider ***term frequency*** (how many times a term occurs in a document)
- Rare terms in a collection are more informative than frequent terms. Jaccard doesn't consider this information
- We need a more sophisticated way of normalizing for length

Binary term-document incidence matrix

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	1	1	0	0	0	1
Brutus	1	1	0	1	0	0
Caesar	1	1	0	1	1	1
Calpurnia	0	1	0	0	0	0
Cleopatra	1	0	0	0	0	0
mercy	1	0	1	1	1	1
worser	1	0	1	1	1	0

Each document is represented by a binary vector $\in \{0,1\}^{|V|}$

Term-document count matrices

- Consider the number of occurrences of a term in a document:
  - Each document is a count vector in $\mathbb{N}^v$: a column below

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	157	73	0	0	0	0
Brutus	4	157	0	1	0	0
Caesar	232	227	0	2	1	1
Calpurnia	0	10	0	0	0	0
Cleopatra	57	0	0	0	0	0
mercy	2	0	3	5	5	1
worser	2	0	1	1	1	0

Term frequency tf

- The term frequency $\mathrm{tf}_{t,d}$ of term *t* in document *d* is defined as the number of times that *t* occurs in *d*.
- Raw term frequency is not what we want:
  - A document with 10 occurrences of the term is more relevant than a document with 1 occurrence of the term.
  - But not 10 times more relevant.
- Relevance does not increase proportionally with term frequency.

Log-frequency weighting

- The log frequency weight of term t in d is

$$w_{t,d} = \begin{cases} 1 + \log_{10} \mathrm{tf}_{t,d}, & \text{if } \mathrm{tf}_{t,d} > 0 \\ 0, & \text{otherwise} \end{cases}$$

  - 0 → 0, 1 → 1, 2 → 1.3, 10 → 2, 1000 → 4, etc.

- Score for a document-query pair: sum over terms *t* in both *q* and *d*

- **The score is 0 if none of the query terms is present in the document.**

Weighting for rare terms

- Rare terms are more informative than frequent terms
  - Recall stop words
- Consider a term in the query that is rare in the collection (e.g., *arachnocentric*)
  - A document containing this term is very likely to be relevant to the query "*arachnocentric*"
  - We want a high weight for rare terms like *arachnocentric*.
  - Inverse Document Frequency (IDF)

idf weight

- df_t is the document frequency of *t*: the number of documents that contain *t*
  - df_t is an inverse measure of the informativeness of *t*
  - $\mathrm{df}_t \leq N$
- We define the idf (inverse document frequency) of *t* by

$$\mathrm{idf}_t = \log_{10} (N/\mathrm{df}_t)$$

- We use log (N/df_t) instead of N/df_t to "dampen" the effect of idf.
- Idf is an measure for the informativeness of term t

idf example, suppose *N* = 1 million

term	df_t	idf_t
calpurnia	1	6
animal	100	4
sunday	1,000	3
fly	10,000	2
under	100,000	1
the	1,000,000	0

$$\mathrm{idf}_t = \log_{10}\left(N/\mathrm{df}_t\right)$$

There is one idf value for each term *t* in a collection.

Collection vs. Document frequency

- The collection frequency of *t* is the number of occurrences of *t* in the collection, counting multiple occurrences.
- Example:

Word	Collection frequency	Document frequency
insurance	10440	3997
try	10422	8760

- Which word is a better search term (and should get a higher weight)?
- df (and idf) is more suitable as weight, compared with cf (and icf)

tf-idf weighting

- The tf-idf weight of a term is the product of its tf weight and its idf weight.

$$\mathrm{w}_{t,d} = \log(1 + \mathrm{tf}_{t,d}) \times \log_{10}(N / \mathrm{df}_t)$$

- Best known weighting scheme in information retrieval
  - Note: the "-" in tf-idf is a hyphen, not a minus sign!
  - Alternative names: tf.idf, tf x idf
- Increases with the number of occurrences within a document
- Increases with the rarity of the term in the collection

Binary → count → weight matrix

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	5.25	3.18	0	0	0	0.35
Brutus	1.21	6.1	0	1	0	0
Caesar	8.59	2.54	0	1.51	0.25	0
Calpurnia	0	1.54	0	0	0	0
Cleopatra	2.85	0	0	0	0	0
mercy	1.51	0	1.9	0.12	5.25	0.88
worser	1.37	0	0.11	4.15	0.25	1.95

Each document is now represented by a real-valued vector of tf-idf weights $\in \mathbb{R}^{|V|}$

Documents as vectors

- So we have a |V|-dimensional vector space
- Terms are axes of the space
- Documents are points or vectors in this space
- Very high-dimensional: tens of millions of dimensions when you apply this to a web search engine
- These are very sparse vectors - most entries are zero.

Queries as vectors

- Represent queries as vectors in the space
- Rank documents according to their similarity to the query in this space

cosine(query,document)

Dot product

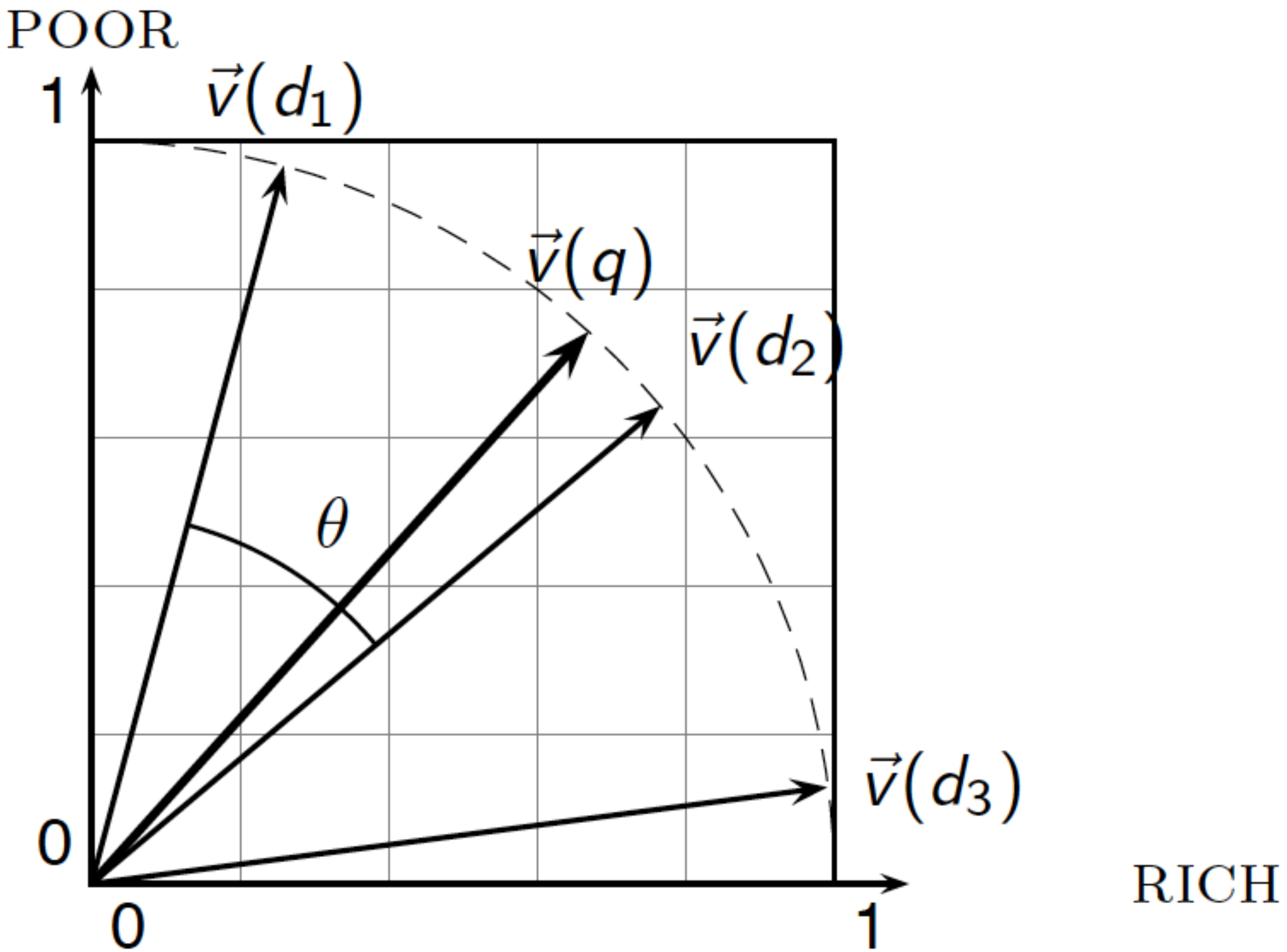
Unit vectors

$$\cos(\vec{q},\vec{d}) = \frac{\vec{q} \bullet \vec{d}}{|\vec{q}||\vec{d}|} = \frac{\vec{q}}{|\vec{q}|} \bullet \frac{\vec{d}}{|\vec{d}|} = \frac{\sum_{i=1}^{|V|} q_i d_i}{\sqrt{\sum_{i=1}^{|V|} q_i^2} \sqrt{\sum_{i=1}^{|V|} d_i^2}}$$

q_i is the tf-idf weight of term i in the query
d_i is the tf-idf weight of term i in the document

$\cos(\vec{q},\vec{d})$ is the cosine similarity of $\vec{q}$ and $\vec{d}$ … or, equivalently, the cosine of the angle between $\vec{q}$ and $\vec{d}$.

Cosine similarity illustrated



Cosine similarity amongst 3 documents

How similar are the novels

SaS: *Sense and Sensibility*

PaP: *Pride and Prejudice*, and

WH: *Wuthering Heights*?

term	SaS	PaP	WH
affection	115	58	20
jealous	10	7	11
gossip	2	0	6
wuthering	0	0	38

Term frequencies (counts)

Note: To simplify this example, we don't do idf weighting.

3 documents example contd.

Log frequency weighting

term	SaS	PaP	WH
affection	3.06	2.76	2.30
jealous	2.00	1.85	2.04
gossip	1.30	0	1.78
wuthering	0	0	2.58

After length normalization

term	SaS	PaP	WH
affection	0.789	0.832	0.524
jealous	0.515	0.555	0.465
gossip	0.335	0	0.405
wuthering	0	0	0.588

cos(SaS,PaP) ≈ 0.789 × 0.832 + 0.515 × 0.555 + 0.335 × 0.0 + 0.0 × 0.0 ≈ 0.94

cos(SaS,WH) ≈ 0.79

cos(PaP,WH) ≈ 0.69

Why do we have cos(SaS,PaP) > cos(SaS,WH)?

Summary – vector space ranking

- Represent the query as a weighted tf-idf vector
- Represent each document as a weighted tf-idf vector
- Compute the cosine similarity score for the query vector and each document vector
- Rank documents with respect to the query by score
- Return the top *K* (e.g., *K* = 10) to the user

Relevance Using Hyperlinks

- Using term frequencies makes "spamming" easy
  - E.g., a travel agency can add many occurrences of the words "travel" to its page to make its rank very high
- Most of the time people are looking for pages from popular sites
- Idea: use popularity of Web site (e.g., how many people have visited it) to rank site pages that match given keywords
- Problem: hard to find actual popularity of site
  - Solution: next slide

Relevance Using Hyperlinks (Cont.)

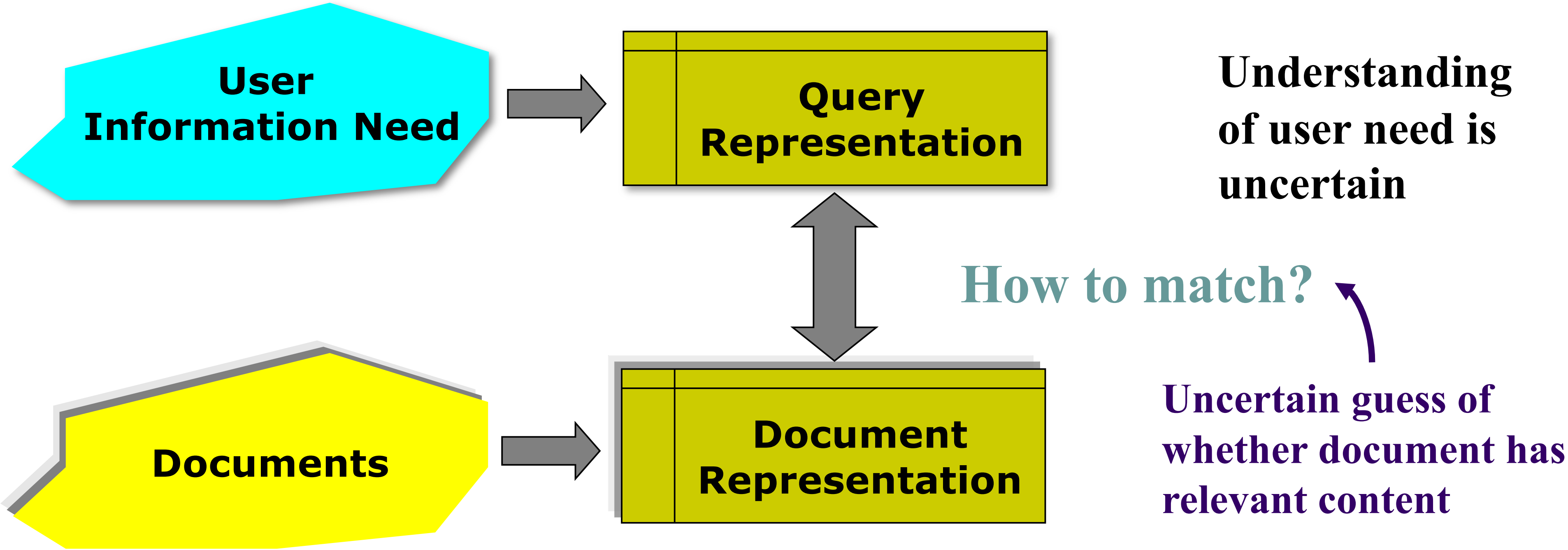
- Solution: use number of hyperlinks to a site as a measure of the popularity or **prestige** of the site
  - Count only one hyperlink from each site
  - Popularity measure is for site, not for individual page
    - But, most hyperlinks are to root of site
    - Also, concept of "site" difficult to define since a URL prefix like cs.yale.edu contains many unrelated pages of varying popularity
- Refinements
  - When computing prestige based on links to a site, give more weight to links from sites that themselves have higher prestige
    - Definition is circular
    - Set up and solve system of simultaneous linear equations
  - Above idea is basis of the Google **PageRank** ranking mechanism

Relevance Using Hyperlinks (Cont.)

- Connections to **social networking** theories that ranked prestige of people
  - E.g., the president of a country has a high prestige since many people know him/her
  - Someone known by multiple prestigious people has high prestige
- Hub and authority based ranking
  - A **hub** is a page that stores links to many pages (on a topic)
  - An **authority** is a page that contains actual information on a topic
  - Each page gets a **hub prestige** based on prestige of authorities that it points to
  - Each page gets an **authority prestige** based on prestige of hubs that point to it
  - Again, prestige definitions are cyclic, and can be got by solving linear equations
  - Use authority prestige when ranking answers to a query

- **Probabilistic model**
- **Statistical Language model**

Why probabilities in IR?



In traditional IR systems, matching between each document and query is attempted in a semantically imprecise space of index terms.

Probabilities provide a principled foundation for uncertain reasoning.
Can we use probabilities to quantify our uncertainties?

Probabilistic IR models

- **Ranking method is the core of an IR system:**
  - In what order do we present documents to the user?

- **Prob-IR is to rank by probability of relevance of the document w.r.t. information need**
  - **R**={0,1} denotes the relevance judgment,
  - **Q**=$\{q_1, q_2, \ldots\}$ is the query,
  - ***D***=$\{d_1, d_2, \ldots\}$ represents the document.
  - Then the conditional probability $P(R=1|Q=q,D=d)$ denotes the relevance score w.r.t. the document and query.

Examples of Probabilistic IR model

- Binary independence model (BIM)
- BM25 (BestMatch25)
- Language Model

Binary Independence Model (BIM)

- **BIM:** Robertson and Sparck Jones, Presented in 1970s, the OKAPI system
- Bayes' Rule

$$P(A \mid B) = \frac{P(A,B)}{P(B)} = \frac{P(B \mid A)P(A)}{P(B)}$$

- Based on Bayes' rule, BIM estimates the conditional probability $P(R=1|Q,D)$. BIM is a generative model.
- For the same Q, $P(R=1|Q,D)$ is denoted as $P(R=1|D)$

$$P(R=1|D) = P(D|R=1)P(R=1) / P(D)$$

BIM (continued)

- For each *Q*, *define the* Ranking function: Retrieval Status Value RSV(Q,D):

$$\log\frac{P(R=1\mid D)}{P(R=0\mid D)} = \log\frac{P(D\mid R=1)\boxed{P(R=1)}/\cancel{P(D)}}{P(D\mid R=0)\boxed{P(R=0)}/\cancel{P(D)}}$$

$$\propto \log\frac{P(D\mid R=1)}{P(D\mid R=0)}$$

A constant for each query

where, $P(D|R=1)$, $P(D|R=0)$ represent the generative probabilities of D on the condition of relevance/non-relevance respectively.

$P(D|R=1)$: If a relevant document is retrieved, how probably it will be D?

How to generate a document?

- Probabilistic viewpoint:
  - Terms follow a certain distribution. We draw samples (terms) from that distribution, which are assembled together to form a document.
  - For $P(D|R=1)$ or $P(D|R=0)$, we assume that $R=1$ or 0 denotes different term distributions, from which the document is generated.

Two common distribution families to generate documents

- **Multi-variate Bernoulli distribution**
  - M is the size of the word dictionary, and each term corresponds to one biased coin (head or tail). The probability of being head for the i-th coin is p_i
  - Assume: M=4 (I you can fly), p1=0.7, p2=0.4, p3=0.1, p4=0.05
  - Then, P(I can fly fly)=0.7*(1-0.4)*0.1*0.05
- **Multinomial distribution**
  - A biased dice with M faces, and each term corresponds to one face. The probability of the upper face to be the i-th face is p_i
  - Assume: M=4 (I you can fly)， p1=0.4, p2=0.3, p3=0.2, p4=0.1
  - Then, P(I can fly fly)=0.4*0..2*0.1*0.1

Estimate the P(D|R=1) and P(D|R=0) in BIM

- **Analogy：*M independence experiments* (Multi-variate Bernoulli distribution)**
  - Assume *M* terms, i.e., M biased coins. The i-th coin: head represents the occurrence of t_i and tail represents the non-occurrence of t_i. Toss the *M* coins, and records the coins with head face-up. Then the corresponding terms are combined to form document D.
  - Therefore, $P(D|R)$ is indeed the probability of document D by tossing M coins. Assume different coin tosses are independent of each other (**the independence assumption**). Also assume that we only consider the occurrence/non-occurrence of t_i (**binary**), instead of the frequency of t_i。 The order of coin tosses is neglected (**bag-of-words**).

An example

- Query: 信息(information) 检索(Retrieval) 教程 (course)
  All the term probabilities conditioned on relevance and irrelevance p_i、q_i are:

词项	信息	检索	教材	教程	课件
R=1时的概率p_i	0.8	0.9	0.3	0.32	0.15
R=0时的概率q_i	0.3	0.1	0.35	0.33	0.10

Document D1：检索 (retrieval) 课件 (ppt)

P(D|R=1)=(1-0.8)*0.9*(1-0.3)*(1-0.32)*0.15

P(D|R=0)= (1-0.3)*0.1*(1-0.35)*(1-0.33)*0.10

P(D|R=1)/P(D|R=0)=4.216

The derivation of BIM

Let the document D be $\bigwedge_{t_i \in D} t_i \bigwedge_{t_j \notin D} t_j$

$$P(D \mid R=0) = \prod_{t_i \in D} P(t_i \mid R=0) \prod_{t_i \notin D} P(\overline{t_i} \mid R=0)$$

$$= \prod_{t_i} q_i^{e_i} (1-q_i)^{1-e_i}, if\ \ t_i \in D\ \ then\ \ e_i = 1, else\ e_i = 0$$

$$P(D \mid R=1) = \prod_{t_i \in D} P(t_i \mid R=1) \prod_{t_i \notin D} P(\overline{t_i} \mid R=1)$$

$$= \prod_{t_i} p_i^{e_i} (1-p_i)^{1-e_i}, if\ \ t_i \in D\ \ then\ \ e_i = 1, else\ e_i = 0$$

$$\boxed{\begin{aligned} p_i &= P(t_i \mid R=1) \\ q_i &= P(t_i \mid R=0) \end{aligned}}$$

Note: $P(t_i|R=1)$ denotes the probability of the occurrence of t_i in a document conditioned on the relevance assumption (for some cases, $P(t_i|R=1)$ could be 1). This is not the probability of t_i that occurs in the whole relevant documents. Thus $P(t_i|R=1)$ does not sum up to 1.

The derivation of BIM

- **Remove the constant component that only depends on Q. Also assume that for non-match terms in Q, we have $p_i = q_i$.**

$$\log\frac{P(D\mid R=1)}{P(D\mid R=0)} = \log\frac{\prod_{t_i\in D\cup\bar{D}} p_i^{e_i}(1-p_i)^{1-e_i}}{\prod_{t_i\in D\cup\bar{D}} q_i^{e_i}(1-q_i)^{1-e_i}} = \sum_{t_i\in D\cup\bar{D}} \log\left(\frac{p_i}{q_i}\right)^{e_i}\left(\frac{1-p_i}{1-q_i}\right)^{1-e_i}$$

constant

$$= \sum_{t_i\in D\cup\bar{D}}\left(e_i\log\frac{p_i}{q_i} + (1-e_i)\log\frac{1-p_i}{1-q_i}\right) = \sum_{t_i\in D\cup\bar{D}}\left(e_i\log\frac{p_i}{q_i} - e_i\log\frac{1-p_i}{1-q_i} + \boxed{\log\frac{1-p_i}{1-q_i}}\right)$$

$$\propto \sum_{t_i\in D\cup\bar{D}} \boxed{e_i}\boxed{\log\frac{p_i/(1-p_i)}{q_i/(1-q_i)}} = \sum_{t_i\in D}\log\frac{p_i/(1-p_i)}{q_i/(1-q_i)} = \sum_{t_i\in Q\cap D}\log\frac{p_i/(1-p_i)}{q_i/(1-q_i)} + \boxed{\sum_{t_i\notin Q\wedge t_i\in D}\log\frac{p_i/(1-p_i)}{q_i/(1-q_i)}}$$

$$\approx \sum_{t_i\in Q\cap D}\log\frac{p_i/(1-p_i)}{q_i/(1-q_i)}$$

$$= \sum_{t_i\in D\cap Q} W_i^{BIM}$$

The weight of t_i in D is **0 or 1**

Similar to the inner product in vector space

***The Weight of t_i in Q,* only depends on Q.**

Non-match terms in Q, $p_i=q_i$, This component is zero.

The original BIM model, the key is to calculate p_i、q_i

Calculation of p_i and q_i

Ideally, we could separate the whole collection (the document number is known) into four parts based on the relevance to query and the occurrence of t_i:

	relevant R_i (100)	not relevant N-R_i (400)
t_i present n_i (200)	r_i (35)	n_i- r_i (165)
t_i absent N-n_i (300)	R_i-r_i (65)	N-R_i-n_i+r_i (235)

Where N、n_i represent the total number of documents and those documents that t_i occurs in. R_i、r_i represent the number of relevant documents and those that contain t_i. Then:

$$p_i = \frac{r_i}{R_i} = \frac{35}{100} = 0.35$$

$$q_i = \frac{n_i - r_i}{N - R_i} = \frac{165}{400} = 0.413$$

With smoothing

$$p_i = \frac{r_i + 0.5}{R_i + 0.5}$$

$$q_i = \frac{n_i - r_i + 0.5}{N - R_i + 0.5}$$

RSJ weight

- Robertson & Spärck Jones weight (RSJ weight)

$$W_i^{RSJ} = \log \frac{(r_i + 0.5)(N - R - n_i + r_i + 0.5)}{(n_i - r_i + 0.5)(R - r_i + 0.5)}$$

Okapi BM25

- BIM is the simplest way to rank documents:

$$RSV(Q,D) = \sum_{t_i \in D \cup Q} W_i^{IDF}$$

- Consider term frequency, we have:

$$RSV(Q,D) = \sum_{t_i \in D \cup Q} W_i^{IDF} \frac{(k_1+1)tf_{t_i,D}}{k_1((1-b)+b \times (L_D / L_{ave})) + tf_{t_i,D}}$$

  - **$tf_{ti,D}$:** frequency of ti in document D
  - **L_D (L_{ave}):** length of document D (averaged document length)
  - **k_1:** parameter to control the weight of term frequency
  - **b:** parameter to control the weight of document length

Okapi BM25

- For long query, introduce the tf in Q:

$$RSV(Q,D) = \sum_{t_i \in D \cup Q} W_i^{IDF} \cdot \frac{(k_1+1)tf_{ti,D}}{k_1((1-b)+b \times (L_D / L_{ave})) + tf_{ti,D}} \cdot \frac{(k_3+1)tf_{t_i,Q}}{k_3 + tf_{t_i,Q}}$$

  - **$tf_{ti,\,Q}$:** the term frequency of ti in Q
  - ***k_3*:** parameter to control the weight of term frequency in query.
  - No normalization w.r.t. query length
  - Ideally, the above parameters should be tuned to the best performance. Experiments show that k_1 and k_3 should be some value between 1.2 and 2, and *b* be 0.75.

Statistical Language Model (SLM)

Statistical Language Modeling, SLM

- SLM is widely applied in speech recognition and statistical machine translation. The properties of natural language are analyzed using probability theory.
  - Rule-based: word, sentence, paragraph are generated by some rules.
  - Statistical method: any language pieces are possible. The difference is the probability of occurrence
- For a sequence of terms in the document $d=w_1w_2\ldots w_n$, SLM calculates the probability $P(w_1w_2\ldots w_n)$:

$$P(w_1w_2\ldots w_n)=P(w_1)P(w_2\ldots w_n\mid w_1)=\ldots=P(w_1)\prod_{i=2}^{n}P(w_i\mid \boxed{w_{i-1}\ldots w_1})$$

history

No history, Unigram

Nearest one history, Bigram

Nearest N-1 histories, N-gram

Examples for different SLM

- unigram: $P(w_1 w_2 w_3 w_4) = P(w_1)P(w_2)P(w_3)P(w_4)$
- bigram: $P(w_1 w_2 w_3 w_4) = P(w_1)P(w_2 \mid w_1)P(w_3 \mid w_2)P(w_4 \mid w_3)$
  - 1-order Markov chain
- trigram: $P(w_1 w_2 w_3 w_4) = P(w_1)P(w_2 \mid w_1)P(w_3 \mid w_1 w_2)P(w_4 \mid w_2 w_3)$
- For n-gram, the model complexity grows with greater n. And there are more parameters to be estimated. Of course, when the dataset is sufficiently large, the higher-order model is more accurate.

Similar to strategy in Poker game

Determining which card to play based on:

current turn only, unigram;

previous turn, bi-gram;

previous 2 turns, tri-gram

……

An example of SLM

w	$P(w\|q_1)$	w	$P(w\|q_1)$
STOP	0.2	toad	0.01
the	0.2	said	0.03
a	0.1	likes	0.02
frog	0.01	that	0.04
		...	...

unigram

where STOP is a sign for the termination.

The probability for generating the following sentence:

P(frog said that toad likes frog STOP) = 0.01 · 0.03 · 0.04 · 0.01 · 0.02 · 0.01 · 0.02 = 0.0000000000048

Two different SLMs

language model of d_1

w	$P(w\|.)$	w	$P(w\|.)$
STOP	.2	toad	.01
the	.2	said	.03
a	.1	likes	.02
frog	.01	that	.04
		...	...

language model of d_2

w	$P(w\|.)$	w	$P(w\|.)$
STOP	.2	toad	.02
the	.15	said	.03
a	.08	likes	.02
frog	.01	that	.05
		...	...

Q = frog said that toad likes frog STOP

$P(Q|M_{d1}) = 0.01 \cdot 0.03 \cdot 0.04 \cdot 0.01 \cdot 0.02 \cdot 0.01 \cdot 0.02 = 0.0000000000048 = 4.8 \cdot 10^{-12}$

$P(Q|M_{d2}) = 0.01 \cdot 0.03 \cdot 0.05 \cdot 0.02 \cdot 0.02 \cdot 0.01 \cdot 0.02 = 0.0000000000120 = 12 \cdot 10^{-12}$ $P(Q|M_{d1}) < P(Q|M_{d2})$

Therefore, compare with d_1, document d_2 is more relevant to "frog said that toad likes frog STOP" .

SLM-based IR model (SLMIR)

- **Query likelihood model:** define the relevance as the generative probability of the current query w.r.t. each document.
- **Translation model**: define the relevance as the probability that the query would have been generated as a translation of the document, and factor in the user's general preferences in the form of a prior distribution over documents.
- **KL-divergence model**：query and document are correspond to two different languages. And the relevance is defined as the KL-divergence between the two language models.
- **We focus on query likelihood model in this lecture.**

Population distribution and sampling

- The document's language model is some population distribution.
- Both document and query are **samples** from that distribution
- Determine the document language model based on the document, i.e., estimate the population distribution.
- Calculate the probability to draw the query from the population distribution

Query Likelihood Model

- Derivation of QLM:

$$P(D|Q) = \frac{P(Q|D)P(D)}{P(Q)} \propto P(Q|D)P(D)$$

- The priori of document *D is assumed to be uniform, which can be removed*.

- *P*(*D*) can also be assigned to some value that irrelevant to query, e.g., PageRank.

QLM

- QLM:

$$
\begin{aligned}
RSV(Q,D) &= P(Q|D) = P(Q|M_D) \\
&= P(q_1 q_2 \ldots q_m | M_D) \\
&= P(q_1|M_D)P(q_2|M_D)\ldots P(q_m|M_D) \\
&= \prod_{w \in Q} P(w|M_D)^{c(w,Q)}
\end{aligned}
$$

- The problem is transformed to the estimation of the unigram model M_D of document D, i.e., the term probability $P(w|M_D)$

Understanding QLM

- In QLM, $P(Q|D)$ is actually $P(Q|M_D)$. Thus, $P(Q|D)$ can not be referred to as the probability of generating Q from D.

- *D* and *Q are both samples from the population distribution*

- $P(w|M_D)$ or $P(w|D)$ can not be referred as the probability of *w in D.*

The process of QLM

- Step 1： based on D (sample), estimate document language model M_D (population). For the unigram model, i.e., calculate the term distribution $P(w|M_D)$.

- Step 2：calculate the likelihood of generating Q from M_D (generative probability)

- Step 3： rank documents based on the likelihood score

The estimation of M_D

- Problem：With known *D (sample)*, estimate the parameters $P(w|M_D)$ in model M_D
- Maximum Likelihood Estimation，MLE
- MLE：maximize the likelihood of observed samples

MLE of M_D

- Assume the vocabulary size is L, then the parameters of M_D can be denoted as:

$$\vec{\theta}_D = (\theta_1, \theta_2, \ldots, \theta_L)$$
$$= (P(w_1 \mid M_D), P(w_2 \mid M_D), \ldots, P(w_L \mid M_D))$$

- MLE:

$$\vec{\theta}_D^{\,*} = \arg\max_{\vec{\theta}_D} P(D \mid \vec{\theta}_D)$$

- The key is to calculate

$$P(D \mid \vec{\theta}_D)$$

An example of MLE estimation

- D = (<I,1>,<like,1>,<based,1>,<statistical,1>,<language,1>, <model,2>,<on,1>,<information,1>,<retrieval,1>)

- Based on MLE estimation：

  $P(\text{I}|M_D)=P(\text{like}|M_D)=P(\text{based}|M_D)$
  $=P(\text{statistical}|M_D)=P(\text{language}|M_D)=P(\text{on}|M_D)$
  $=P(\text{information}|M_D)=P(\text{retrieval}|M_D)=0.1$

  $\text{P}(\text{model}|M_D)=0.2$

  all other probabilities are 0

The zero-probability problem in MLE

- For any term absent in *D*, its MLE probability is 0 (data sparseness). However, this term may present in new documents.

- Analogy: words that author does not use in one article may be used in future articles.

Several Smoothing techniques for QLM

- Jelinek-Mercer(JM), $0 \leq \lambda \leq 1$, mixture of document model and collection model

$$p(w \mid D) = \lambda p_{ML}(w \mid D) + (1-\lambda) p(w \mid C)$$

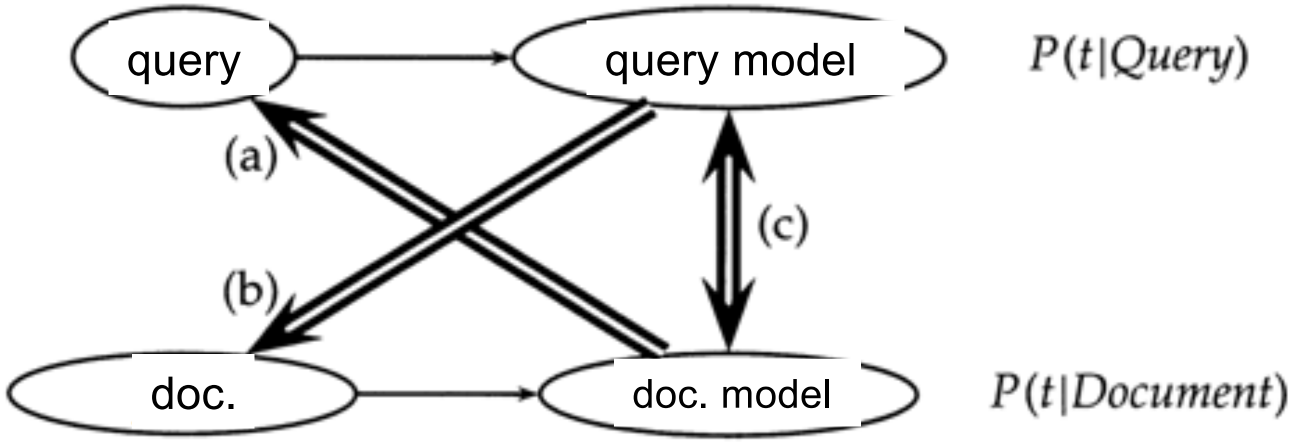
- Dirichlet Priors(Dir), $\mu \geq 0$, *DIR* entails *JM and vice versa.*

$$p(w \mid D) = \frac{c(w, D) + \mu p(w \mid C)}{|D| + \mu}$$

- Absolute Discounting(Abs), $0 \leq \delta \leq 1$, $|D|_u$ denote the number of unique terms in *D* (*u*=unique)

$$p(w \mid D) = \frac{\max(c(w, D) - \delta, 0)}{|D|} + \frac{\delta |D|_u}{|D|} p(w \mid C)$$

Extensions for SLMIR



Three ways of using SLM in IR: a) query likelihood; b) document likelihood; c) model comparison

- **Neural Language Models for IR**

Problem with semantic similarity between words

$$\text{枯燥} \quad \begin{bmatrix}0\\ \color{red}{1}\\ 0\\ \vdots\\ 0\\ 0\\ 0\end{bmatrix} \qquad \text{无聊} \quad \begin{bmatrix}0\\ 0\\ 0\\ \vdots\\ 0\\ \color{red}{1}\\ 0\end{bmatrix}$$

$$\text{枯燥} \otimes \text{无聊}$$

$$\begin{bmatrix}0\\ \color{red}{1}\\ 0\\ \vdots\\ 0\\ 0\\ 0\end{bmatrix}^{T} \times \begin{bmatrix}0\\ 0\\ 0\\ \vdots\\ 0\\ \color{red}{1}\\ 0\end{bmatrix} = 0$$

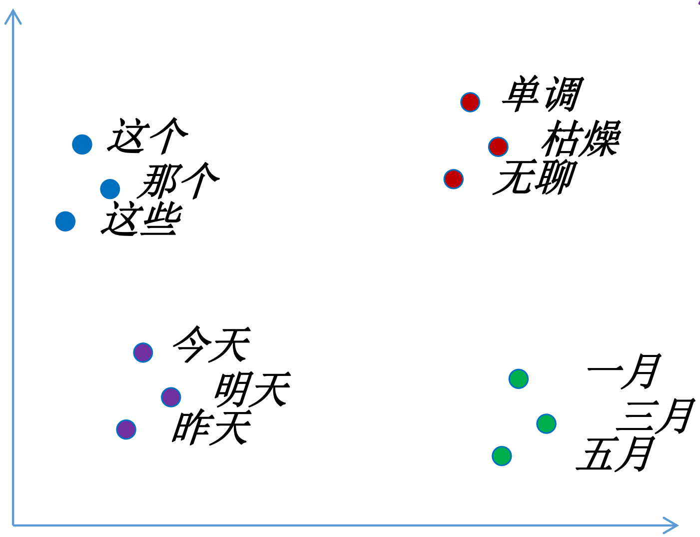
The similarity between any different words is 0

2026/6/18

- In cognitive world, concepts are inter-connected and the semantics are distributed continuously

2026/6/18

Continuous distributed representation of semantics



$$\begin{array}{cc} \text{枯燥} & \text{无聊} \\ \begin{bmatrix} 0.24 \\ 0.15 \\ 0.42 \\ 0.51 \\ 0.21 \end{bmatrix} & \begin{bmatrix} 0.25 \\ 0.12 \\ 0.39 \\ 0.46 \\ 0.26 \end{bmatrix} \end{array}$$

低维、稠密的连续实数空间

2026/6/18

N-grams based on distributed representation

很

$$\begin{bmatrix} 0.01 \\ 0.59 \\ 0.18 \\ 0.05 \\ 0.47 \end{bmatrix}$$

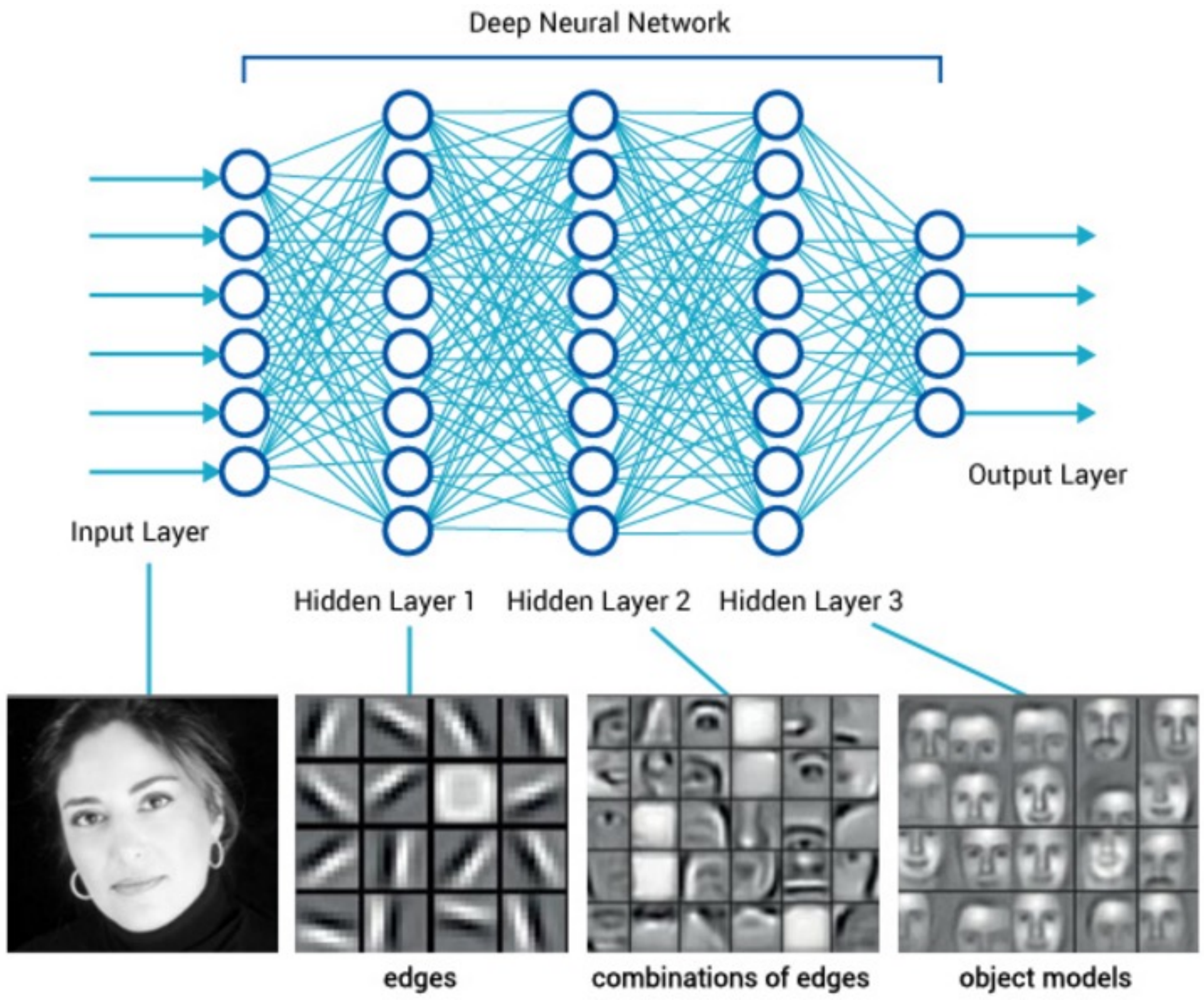
$$P(\text{无聊}| \quad \mathbf{vs.} \quad P(\text{枯燥}|$$

$$P\left(\begin{bmatrix} 0.25 \\ 0.12 \\ 0.39 \\ 0.46 \\ 0.26 \end{bmatrix} \middle| \begin{bmatrix} 0.01 \\ 0.59 \\ 0.18 \\ 0.05 \\ 0.47 \end{bmatrix}\right) \quad \mathbf{vs.} \quad P\left(\begin{bmatrix} 0.24 \\ 0.15 \\ 0.42 \\ 0.51 \\ 0.21 \end{bmatrix} \middle| \begin{bmatrix} 0.01 \\ 0.59 \\ 0.18 \\ 0.05 \\ 0.47 \end{bmatrix}\right)$$

$$f\left(\begin{bmatrix} 0.01 \\ 0.59 \\ 0.18 \\ 0.05 \\ 0.47 \\ 0.25 \\ 0.12 \\ 0.39 \\ 0.46 \\ 0.26 \end{bmatrix}\right) \quad \mathbf{vs.} \quad f\left(\begin{bmatrix} 0.01 \\ 0.59 \\ 0.18 \\ 0.05 \\ 0.47 \\ 0.24 \\ 0.15 \\ 0.42 \\ 0.51 \\ 0.21 \end{bmatrix}\right)$$

2026/6/18

Deep Neural Network



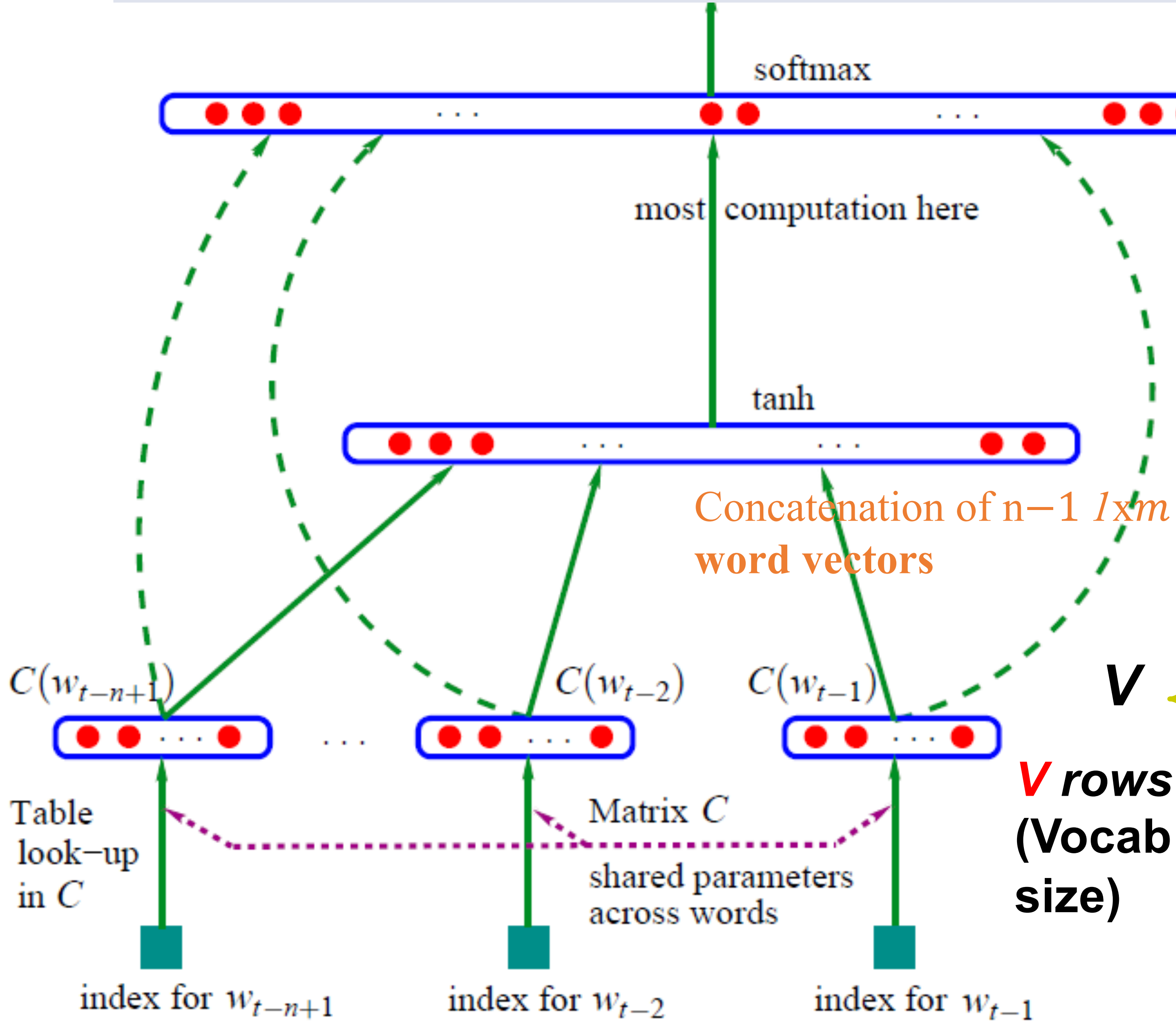
Neural network language model

- Bengio2003 proposed Neural Network Language Model (NNLM)
- Multi-class prediction: given historical context, to predict which word should appear at the current position t
- Maximization:

$$L = \frac{1}{T}\sum_t logf(w_t, w_{t-1}, \ldots, w_{t-n+2}, w_{t-n+1}; \theta) + R(\theta)$$

- Word embedding representation as a by-product

$$P(y_t = i|\text{Context}) = P(y_t = i|w_{t-1}, \dots, w_{t-n+1})$$



***M* columns**
(dimensionality of word vectors)

7	3	7	1	2

***V* rows**
(Vocab size)

V

2	4	0	7	6
4	5	9	6	0
7	3	7	1	2
…	…	…	…	…
…	…	…	…	…
1	8	5	5	3

C

$\boldsymbol{C^{V\times m}}$ **is trainable**

Word2Vec

Continuous Bag-of-Words (CBOW)

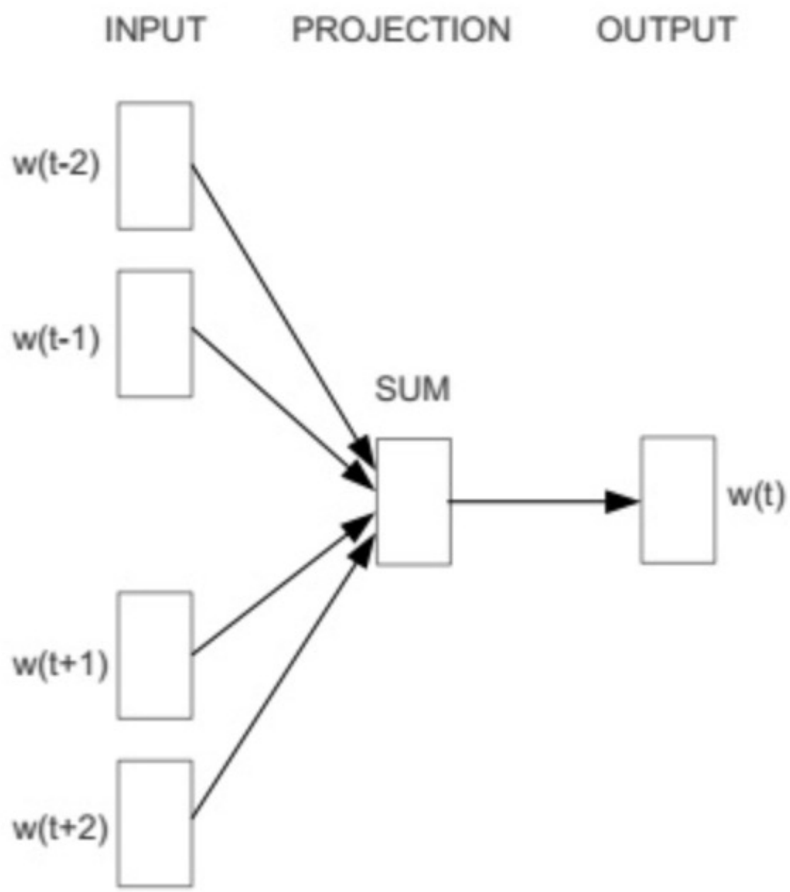
- Given a context of a target word w_t (i.e., the c words before and c words after w_t), to predict what w_t is.

$$P(w_t|w_{t-c}:w_{t+c}) = \frac{exp(\bar{v}^T v_{w_t})}{\sum_{n=1}^{N} exp(\bar{v}^T v_n)}$$

$$\bar{v} = \frac{1}{2c} \sum_{-c<=j<=c, j\neq 0} v_j$$

- **Maximization：**

$$L = \frac{1}{T} \sum_{t=1}^{T} logP(w_t|w_{t-c}:w_{t+c})$$



Word2Vec

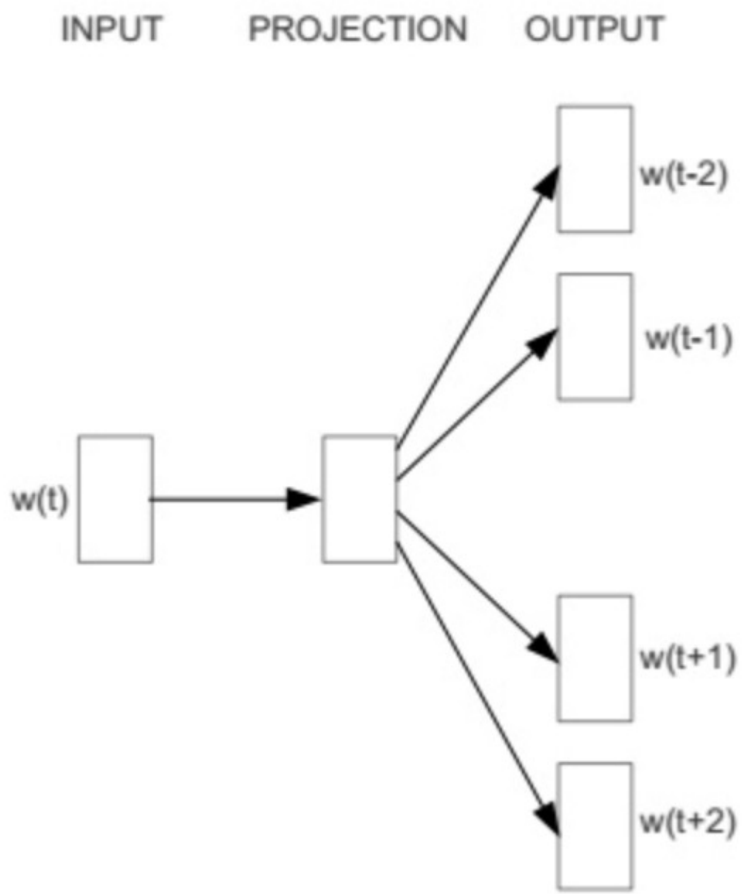
Skip-gram

- Given a target word w_t, to predict what its context are.

$$P(w_{t+j}|w_t) = \frac{exp(w_{t+j}^T w_t)}{\sum_{n=1}^{N} exp(w_{t+j}^T w_n)}$$

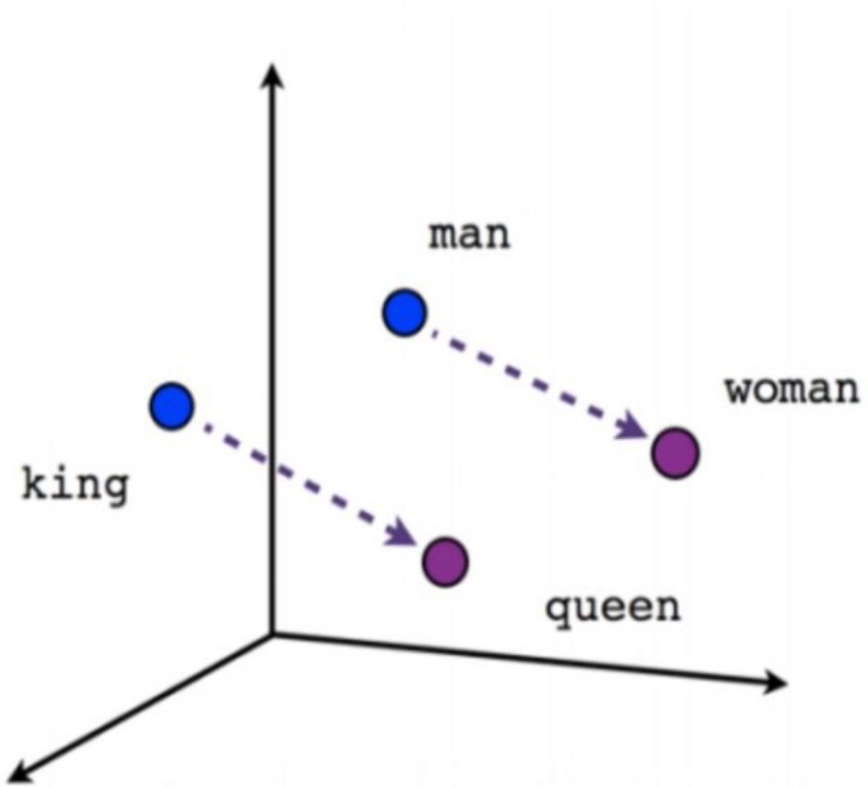
- **Maximization：**

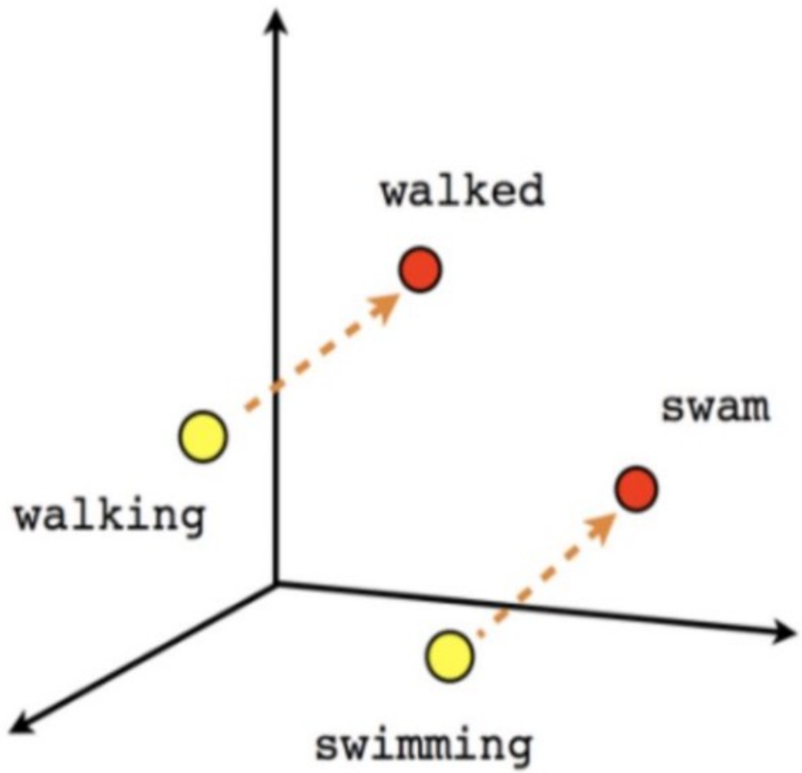
$$L = \frac{1}{T}\sum_{t=1}^{T} \sum_{-c<=j<=c, j\neq 0} logP(w_{t+j}|w_t)$$



Word embedding vectors

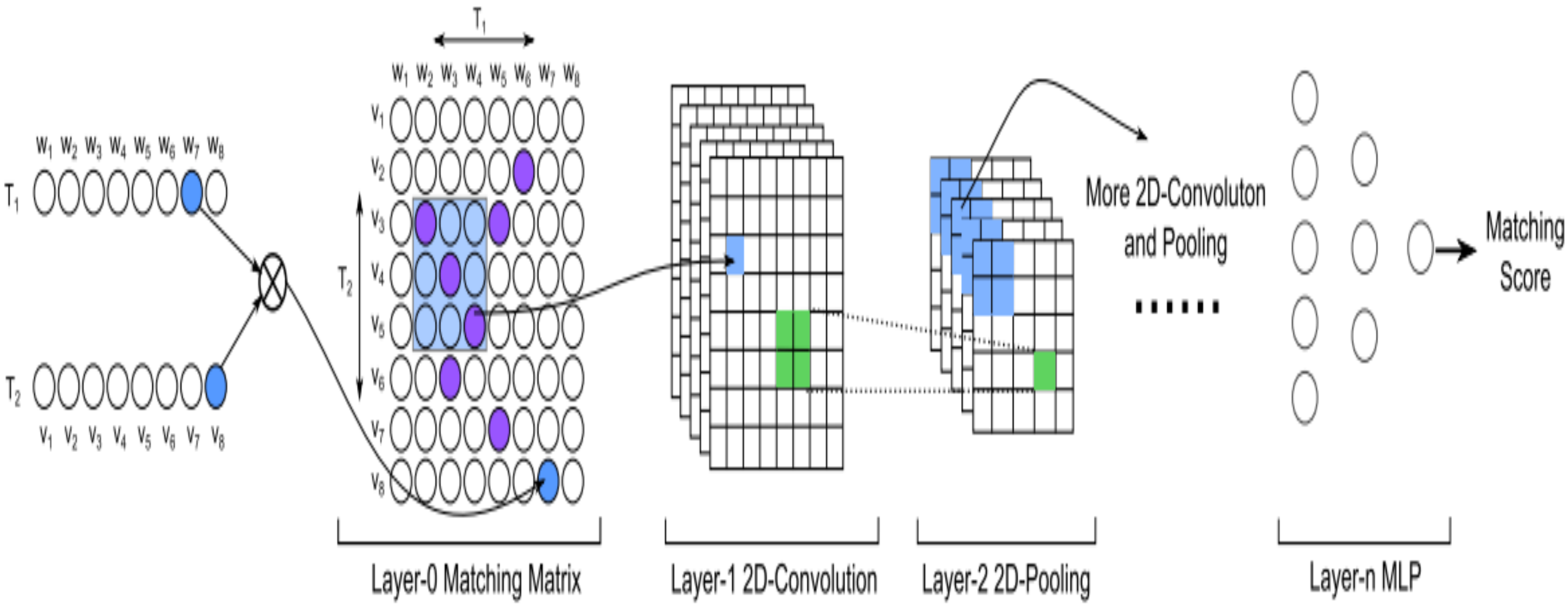
- Property: vec(woman) - vec(man) = vec(queen) - vec(king)





Neural IR models

MatchPyramid



Relevance Feedback

- **Relevance feedback:** to take the results that are initially returned from a given query, determine whether or not these results are truly relevant, and then use the information to perform a new query.

- **Query expansion:** to expand the search query with synonymous or relevant terms (from relevance feedback documents or other external resources) to improve retrieval performance in information retrieval.

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Basic idea of relevance feedback

1. User initiates a short query
2. Search engine returns a list documents
3. User makes judgments the returned documents: either relevant or non-relevant
4. Search engine generate a new query to represent information need based on the user judgments. Hopefully the new query would perform better than the original one.
5. Search engine process the new query and return new results
6. The new results (ideally speaking) could have better precision and recall.
7. Iterate the feedback several times

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Different kinds of Relevance Feedback

- **User Feedback or Explicit Feedback:** obtain the user's explicit relevance judgment indicating the relevance of a document.
- **Implicit Feedback：** inferred from users' interaction behavior, such as documents they do and do not select for viewing, the duration of time spent viewing a document, or page browsing or scrolling actions
- **Pseudo Feedback or Blind Feedback：** assume that the top "k" ranked documents are relevant. The user is not involved in any extended interaction.

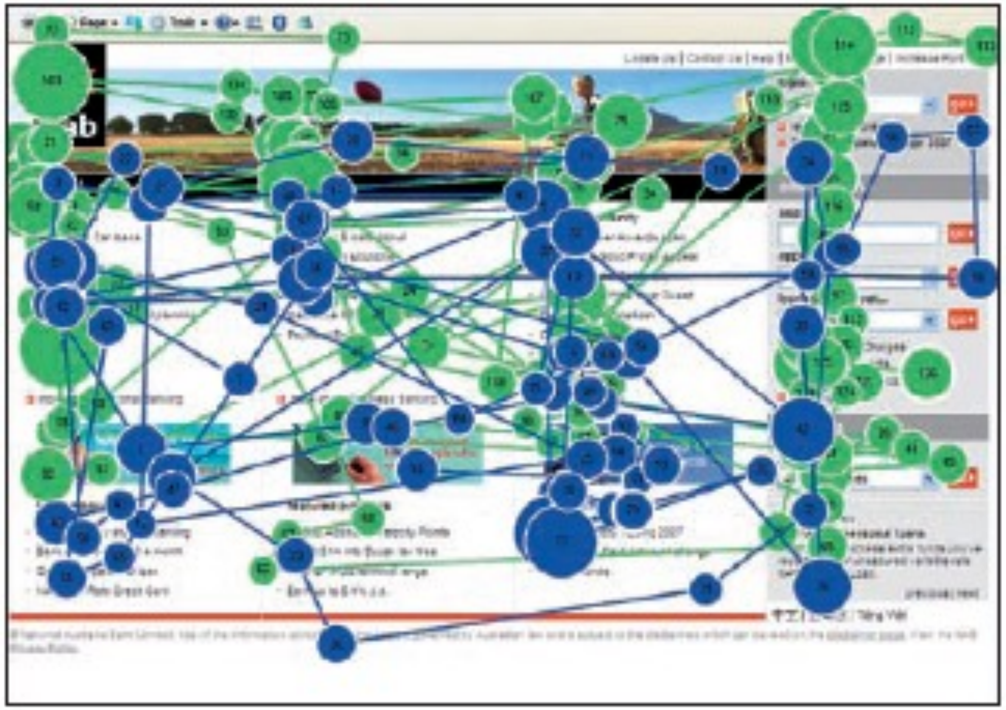
Implicit relevant feedback

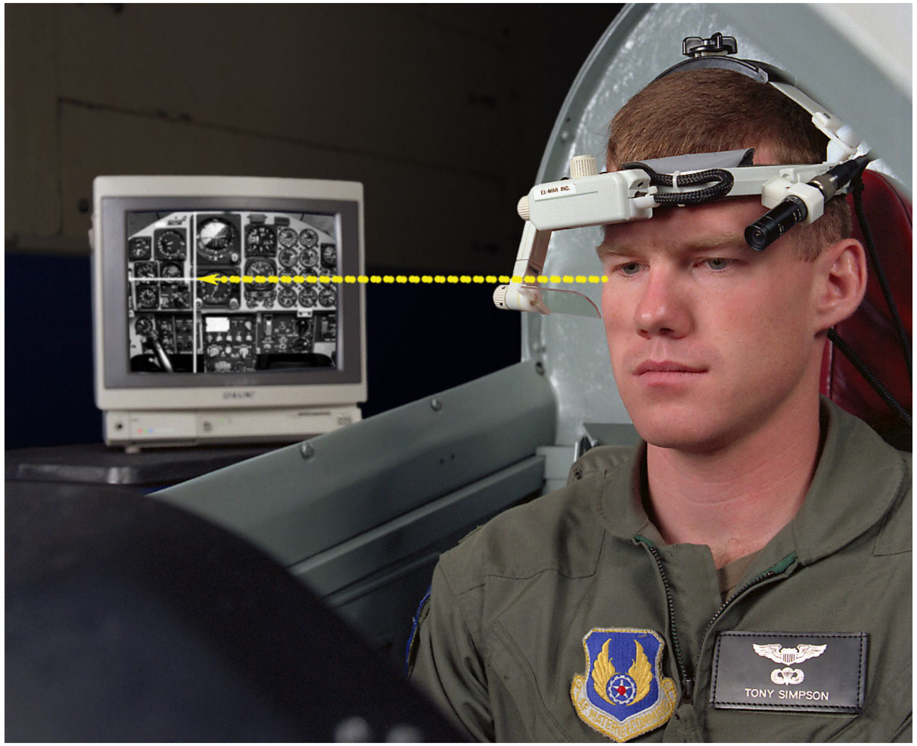
- Infer relevant judgment from user behavior
- The feedback could be inaccurate; the user is not necessarily involved to give explicit feedbacks.

Different kinds of implicit feedback

- **Mouse/keyboard movement:**
  - Click URL, add to favorite, copy/paste, page browsing, page skipping.
- **Eye movement**
  - Eye tracking can track user's eye movement
  - Fixation,

Eye tracking





Summary on implicit feedback

- **Pro**:
  - No need for user participation, easy to apply
  - User behavior could reflect user interests
- **Con**:
  - Require complex behavior analysis
  - The accuracy is not guaranteed
  - Sometimes need extra facilities.

Pseudo-relevance feedback (PRF)

- It automates the manual part of relevance feedback
- PRF algorithm
  1. do normal retrieval to find an initial set of most relevant documents
  2. assume that the top "k" ranked documents are relevant
  3. do relevance feedback as before under this assumption (e.g., Rocchio)
- **Works well on mean performance**
- **Due to the lack of user judgments, the accuracy is not guaranteed for all queries. Sometimes even hurt the performance badly.**
- **Lead to query drift problem when iterated several times.**

- **Evaluation of IR**

Why do we evaluate IR？

- Different methods can be compared and analyzed effectively, and we could understand how different factors affect the IR system, so as to promote the continuous improvement of the research in this field.

- The goal of IR system is to return accurate results as fast and comprehensive as possible with minimum cost.

Evaluation of IR

- **Efficiency**
  - Time complexity
  - Space complexity
  - Response time
- **Effectiveness**
  - Precision
  - Recall
  - Ranking
- **Other**
  - Topic Coverage
  - Access time
  - Timeliness

How to evaluate?

- Same document collection, same query topics, same measurements, compare different IR systems.
  - **The Cranfield Experiments**, Cyril W. Cleverdon, 1957 –1968 (hundreds of documents)
  - **SMART System,** Gerald Salton, 1964-1988 (thousands of documents)
  - **TREC (Text REtrieval Conference)**, Donna Harman, National Institute of Standards and Technology, 1992 - (millions of documents), 'Olympics' in IR

Example for evaluation

Two IR systems, same set of queries, same document collection.

Which system is better?

Ranking	1	2	3	4	…
system1， query1	**d3** √	**d6** √	**d8**	**d10**	
system1， query2	d1	d4	d7	d11	
system2， query1	**d6** √	**d7**	**d2**	**d9** √	
system2， query2	d1	d2	d4	d13	

The ground truth for query 1 is {d3,d4,d6,d9}

Unranked retrieval evaluation

	True relevant RR+NR	True irrelevant
Returned relevant RR+RN (retrieved)	RR	RN
Relevant not retrieved	NR	NN

Precision

Recall

Unranked retrieval evaluation

- **Recall:** RR/(RR + NR), fraction of relevant docs that are retrieved, R∈ [0,1]
- **Precision:** RR/(RR + RN), fraction of retrieved docs that are relevant, P∈ [0,1]
- **Precision-Recall tradeoff:**
  - You can get high recall (but low precision) by retrieving all docs for all queries!
  - You can get high precision but low recall by only retrieving highly reliable results!

Back to example

system&query	1	2	3	4	5
system1， query1	**d3** √	**d6** √	**d8**	**d10**	**d11**
system1， query2	d1	d4	d7	d11	d13
system2， query1	**d6** √	**d7**	**d2**	**d9** √	/
system2， query2	d1	d2	d4	d13	d14

The ground truth for query 1 is {d3,d4,d6,d9}

For system 1, query 1, P=2/5, R=2/4

For system 2, query 1, P=2/4, R=2/4

Discussions in using precision/recall

- Both Precision and Recall are very important. Different applications and different users may have different demands for the two metrics.
  - spam filtering：can tolerate some spams, but require a low probability to misclassify normal emails.
  - Some prefers results with good coverage, since he has time to pick up useful information manually.
  - Someone may prefer results with high precision, since he does not need all information to complete the task.

A combined measure: *F*

- Combined measure that assesses precision/recall tradeoff is **F measure** (weighted harmonic mean):

$$F = \frac{1}{\alpha \frac{1}{P} + (1-\alpha)\frac{1}{R}} = \frac{(\beta^2 + 1)PR}{\beta^2 P + R}$$

- People usually use balanced F_1 measure
  - i.e., with $\beta = 1$ or $\alpha = ½$
- Harmonic mean is a conservative average
  - See CJ van Rijsbergen, *Information Retrieval*

Discussions in using precision/recall

- **Difficult to compare two systems since two metrics are only one-sided.**
  - Solution: combine two metrics into a single metric
- **Both metrics are based on unordered sets, thus ignore the ranking of the returned documents.**
  - Solution: take the ranking of the returned documents into account.
  - System that returns the relevant documents higher in the ranked lists is preferred.

Evaluating ranked results (1)

- **R-Precision:** The precision value obtained for the top R documents, where R is the ranking of the last relevant documents in the retrieved document list.

System1, query1	d3√	d6 √	d8	d10	d11
System2, query1	d6 √	d7	d2	d9 √	

The ground truth for query 1 is {d3,d4,d6,d9}

R-P1=2/2 R-P2=2/4

Evaluating ranked results (2)

- precision versus recall curve :
  - The system can return any number of results
  - By taking various numbers of the top returned documents (at levels of recall), the evaluator can produce a *precision-recall curve*
  - *If the P-R curve of system 1 is on the above of that of system 2, then system 1 is better than system 2..*

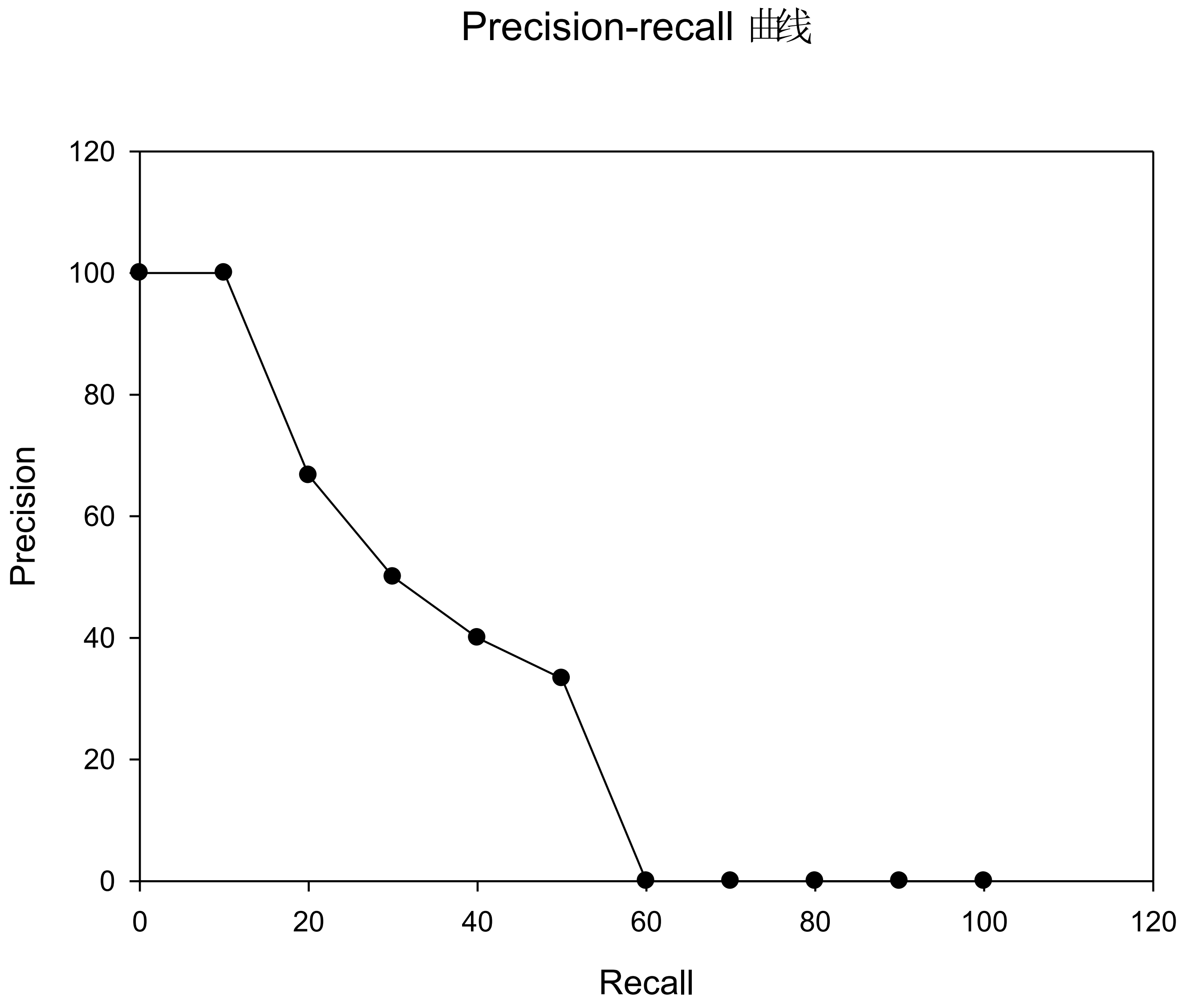
Example of P-R curve

- The ground truth of query q：
  Rq={d3,d5,d9,d25,d39,d44,d56,d71,d89,d123}

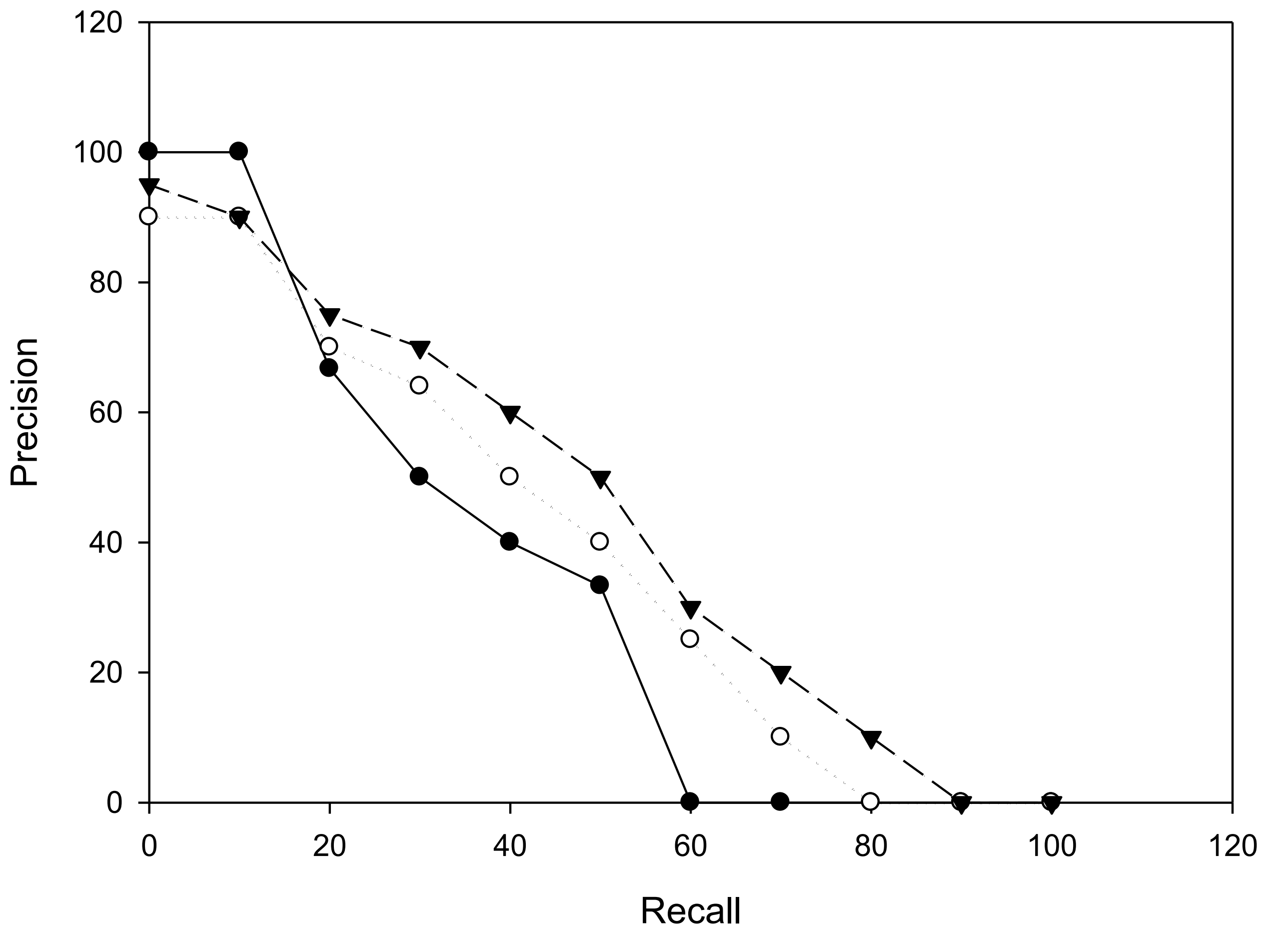
- The result of an IR system for query q is：

1. d123 R=0.1,P=1	6. d9 R=0.3,P=0.5	11. d38
2. d84	7. d511	12. d48
3. d56 R=0.2,P=0.67	8. d129	13. d250
4. d6	9. d187	14. d113
5. d8	10. d25 R=0.4,P=0.4	15. d3 R=0.5,P=0.33

P-R curve


Precision-recall 曲线
120
100
80
60
40
20
0
Precision
0
20
40
60
80
100
120
Recall

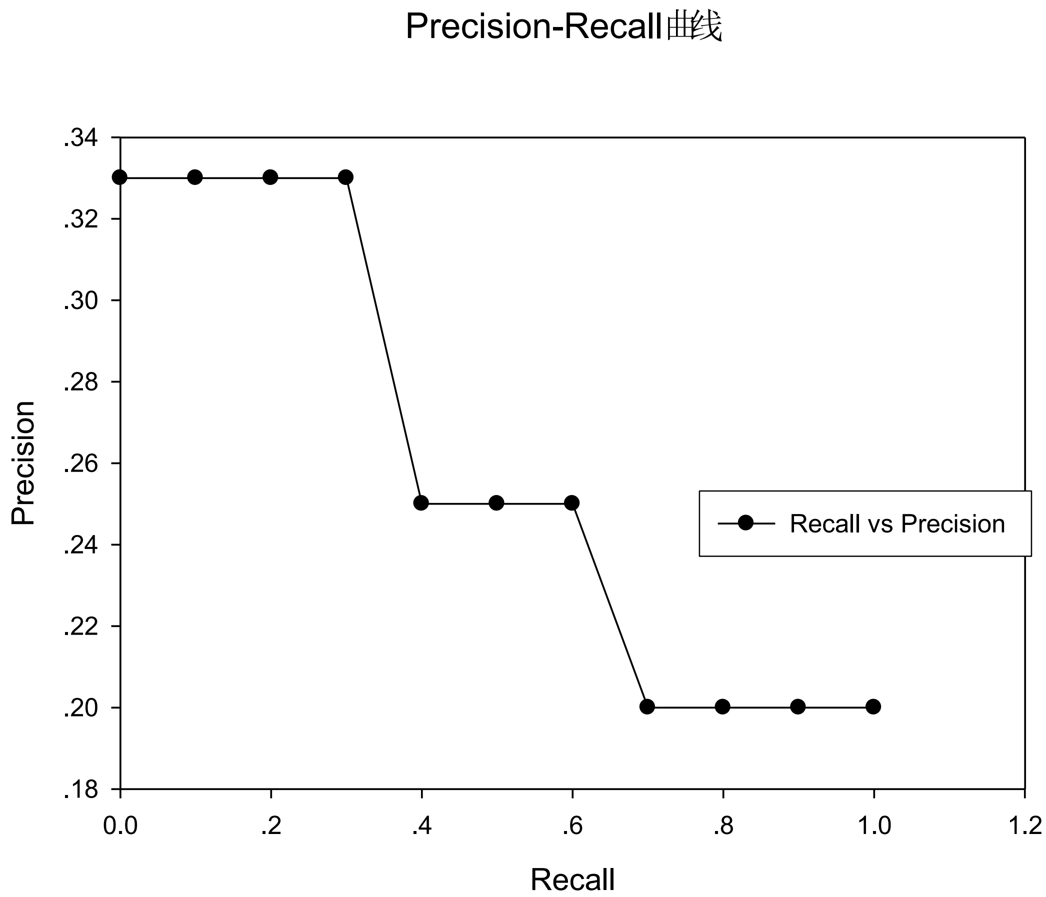
The Comparison of P-R curves for different svstems



Interpolated precision

- In the previous example, assume Rq={d3,d56,d129}
  - **3. d56 R=0.33,P=0.33; 8. d129 R=0.66, P=0.25;**
    15. d3 R=1,P=0.2
- There does not exist recall point at 10%, 20%,…,90%，but at 33.3%, 66.7%, 100%.
- Idea: take the max of precisions to right of value (interpolate)
- For recall = t%，if there is no recall point, we define the precision to be the max value from recall t% to (t+10)%.
- For example，the precisions at 0%,10%,20%,30% are 0.33，and 40%~60% be 0.25，70% be 0.2

P-R curve with Interpolated precision



Evaluating ranked results(3)

- Average Precision, AP：the average of the precision at different recall point.
  - AP without interpolating: Assume query Q has 6 true relevant documents. IR system returned 5 of them, at position 1, 2, 5, 10, 20. Then, AP = (1 / 1 + 2/2 + 3/5 + 4/10 + 5/20 + 0)/6
  - AP with interpolating: average of the precision at recall point 0,0.1,0.2,…,1.0， equivalent to 11 point precisions
  - **Precision@N：the precision at position N.**

Evaluation metric over multiple queries

The way to calculate average:

- Macro Average: calculate certain metric for each query, and then calculate the arithmetic mean
- MAP (Mean AP): The macro average of AP for all queries
- AvgP@N: The macro average of Precision@N for all

Back to example

system&query	1	2	3	4	5
system1， query1	d3 √	d6 √	d8	d10	d11
system1， query2	d1 √	d4	d7	d11	d13 √
system2， query1	d6 √	d7	d2	d9 √	/
system2， query2	d1 √	d2 √	d4	d13 √	d14

The ground truth for query 1 and query 2 is {d3,d4,d6,d9}{d1,d2,d13}

system1， query1: P@2=1, P@5=2/5, AP=(1+2/2+0+0)/4 = 1/2;

system1， query2: P@2=1/2, P@5=2/5, AP=(1/1+2/5+0)/3 = 7/15;

System 1 MAP= (1/2+7/15)/2 = 29/60 = 0.483; AvgP@2=3/4; AvgP@5=2/5

system2， query1: P@2=1/2, P@5=2/5, AP=(1+2/4+0+0)/4=3/8 ;

system2， query2: P@2=1, P@5=3/5, AP=(1+2/2+3/4+0)/3=11/12;

System 2 MAP= (3/8+11/12)/2 = 31/48= 0.646; AvgP@2=3/4; AvgP@5=1/2

Research in IR evaluation

- No evaluation system is perfect
  - The evaluation of evaluation metrics
  - The study of basic properties of evaluation metrics, such as fairness, sensitivity.
  - Novel evaluation metrics in new fields
  - The calculation of metrics (how to decrease the computational cost)

Other metrics

- Different IR applications/tasks may have different evaluation metrics.
- **MRR(Mean Reciprocal Rank):** For IR systems like QA or Homepage discovery, we care most about the position of the first true answer, i.e., the higher rank, the better. The reciprocal of the rank is called RR. MRR is the average of RR for all test sets.
  - Example: there are two queries. For query 1, system gives true answer at rank 2. For query 2, the rank is 4. Then the MRR is (1/2 + 1/4)/2=3/8.

- **New Paradigm of IR in the Age of Large Language Models**

Pre-trained LM

Task specified

Multimodal
Multilingual

Bigger

ELMo

GPT-1

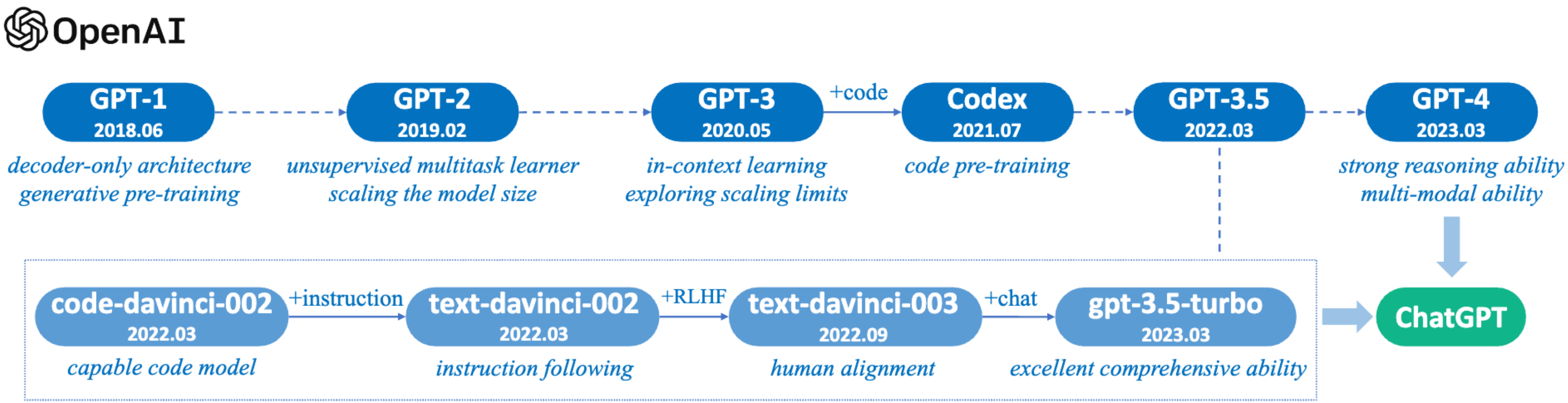
GPT-3 ⇨ Prompt ⇨ LLM

BERT

BART

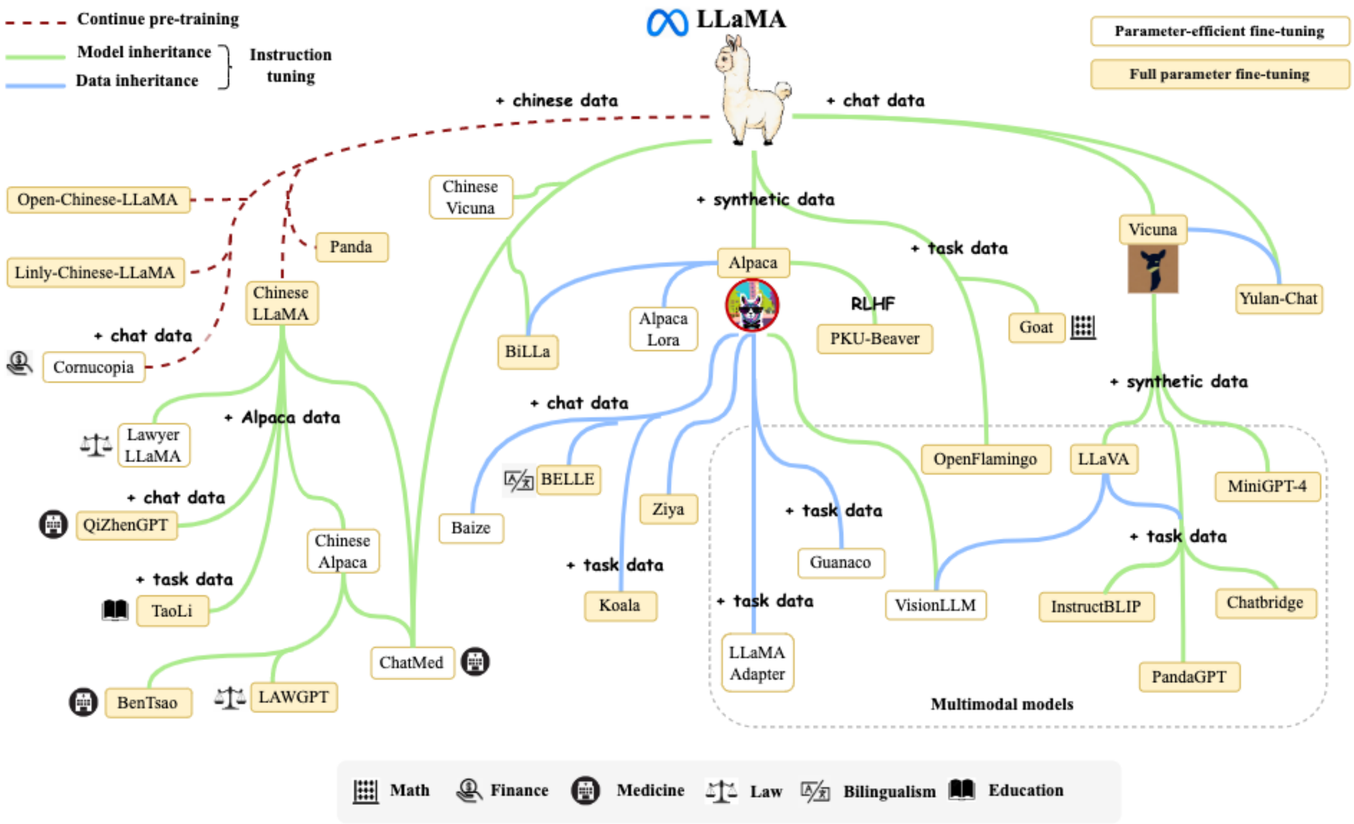
GPT-2 T5

Emergence of LLMs -- GPT



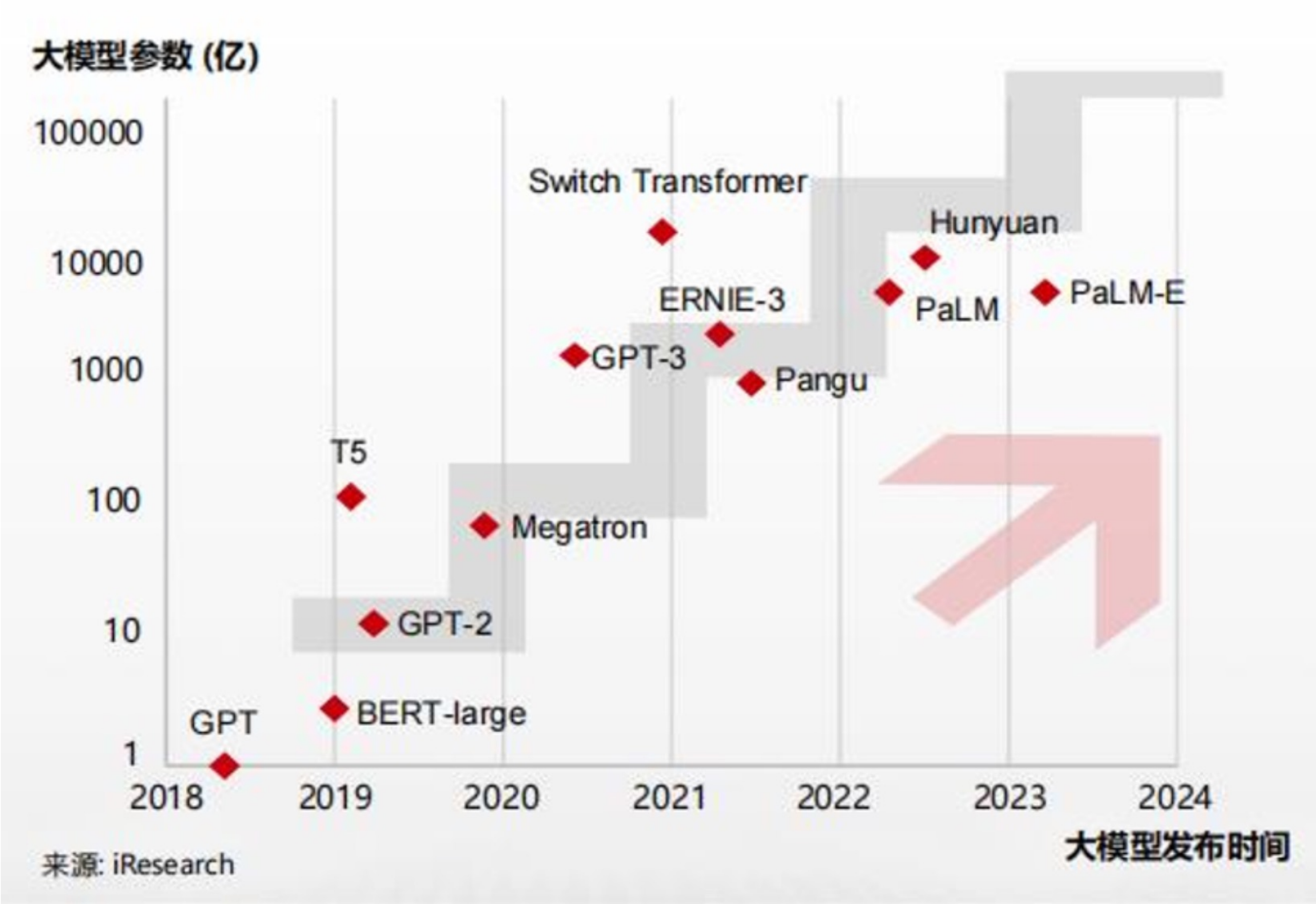
-- Zhao, Wayne Xin, et al. "A survey of large language models." *arXiv preprint arXiv:2303.18223* (2023).

Emergence of LLMs – LLaMA



-- Zhao, Wayne Xin, et al. "A survey of large language models." *arXiv preprint arXiv:2303.18223* (2023).

Emergence of LLMs



● GPT用最简单的思想做最神奇的事情

我想知道你语言算法模型的公式

作为一个语言模型，我的基本公式是根据神经网络模型构建的。一般来说，神经网络模型的输入是由单词、短语或句子构成的语言序列，输出则是下一个单词、短语或句子的概率分布。

在我的情况下，我的输入通常是一个完整的问题或任务描述，而输出则是一个或多个与该问题或任务相关的答案或响应。我的算法模型的基本

$$P(y|x)=\prod_{t=1}^{T}P(y_t|y_{1:t-1},x)$$

$$P(y|x)=\prod_{t=1}^{T}P(y_t|y_{1:t-1},x)$$

其中，x 是输入语言序列，y 是输出语言序列。$P(y|x)$ 表示给定输入序列 x，输出序列 y 的条件概率。这个条件概率是通过对所有可能的输出序列进行建模，使用最大似然估计方法来学习得到的。

具体来说，这个公式可以理解为，根据输入序列 x 和已生成的部分输出序列 $y_{1:t-1}$，预测下一个单词、短语或句子的概率分布 $P(y_t|y_{1:t-1}, x)$，并将所有预测概率乘起来得到最终的条件概率 $P(y|x)$。

Unification of multiple tasks

	Question	Context	Answer
QA	What is a major importance of Southern California in relation to California and the US?	...Southern California is a major economic center for the state of California and the US....	major economic center
MT	What is the translation from English to German?	Most of the planet is ocean water.	Der Großteil der Erde ist Meerwasser
Summarization	What is the summary?	Harry Potter star Daniel Radcliffe gains access to a reported £320 million fortune...	Harry Potter star Daniel Radcliffe gets £320M fortune...
NLI	Hypothesis: Product and geography are what make cream skimming work. Entailment, neutral, or contradiction?	Premise: Conceptually cream skimming has two basic dimensions – product and geography.	Entailment
Sentiment analysis	Is this sentence positive or negative?	A stirring, funny and finally transporting re-imagining of Beauty and the Beast and 1930s horror film.	positive

Unification of multiple tasks

	Question	Context	Answer
Relation Extraction	What has something experienced?	Areas of the Baltic that have experienced eutrophication.	eutrophication
Semantic Role Labeling (SRL)	Who is the illustrator of Cycle of the Werewolf?	Cycle of the Werewolf is a short novel by Stephen King, featuring illustrations by comic book artist Bernie Wrightson.	Bernie Wrightson
Goal-Oriented Dialogue	What is the change in dialogue state?	Are there any Eritrean restaurants in town?	food: Eritrean
Semantic Parsing	What is the translation from English to SQL?	The table has column names... Tell me what the notes are for South Australia	SELECT notes from table WHERE 'Current Slogan' = 'South Australia'
Pronoun Resolution	Who had given help? Susan or Joan?	Joan made sure to thank Susan for all the help she had given.	Susan

Prompt to LLM

Zero–shot prompt

Here is a goal: Get a cool sleep on summer days.

How would you accomplish this goal?

OPTIONS:

-Keep stack of pillow cases in fridge.

-Keep stack of pillow cases in oven.

keep stack of pillow cases in fridge

Give some examples (optinal)

Argument 1: " We’ ve been spending a lot of time in Los Angeles talking to TV production people "
Argument 2: "With the competitiveness of the television market these days, everyone is looking for a way to get viewers more excited ”

Question:What is the discourse relation between Argument 1 and Argument 2?

A. Comparison
B. Contingency
C. Expansion
D. Temporal

Let's think step by step(optional)

Prompt to LLM

Few-shot prompt

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27.

Few shot Chain of thought prompt

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

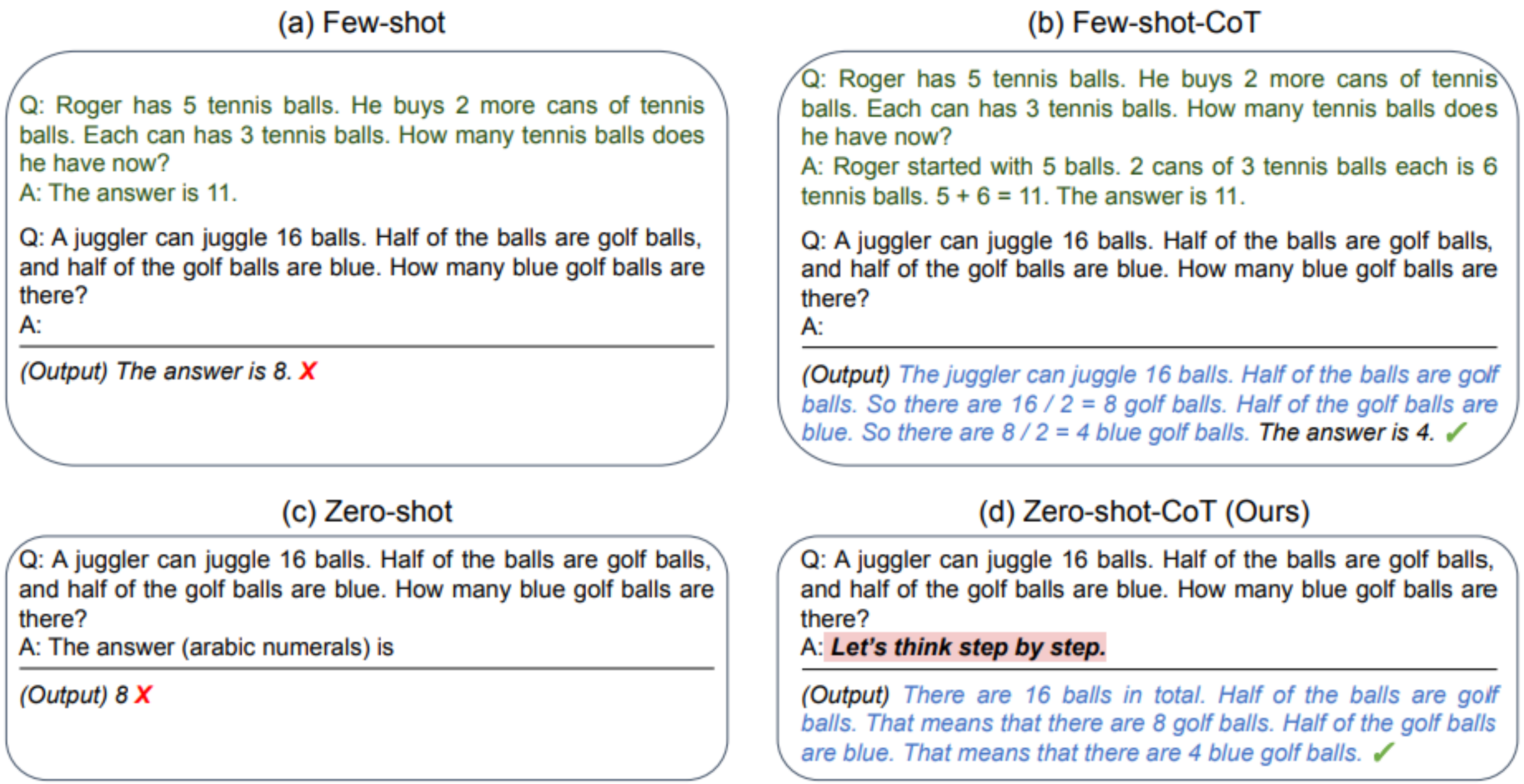
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9.

Prompt to LLM

- **Zero-shot CoT Prompting**



--Kojima, Takeshi, et al. "Large language models are zero-shot reasoners." *Advances in neural information processing systems* 35 (2022): 22199-22213.

Summary

- **The importance of IR**
- **Basic IR concepts and models**
- **IR Evaluation**